

# Supplementary Material for “Faster than Light: Latency Measurement Errors in Message-Broker Benchmarks”

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This document is the supplementary material for the paper named above, and it follows that paper’s order and its names. Table S1 lists, for each section of the paper, the sections here that hold its evidence. Table S2 defines the terms both documents use, each with one meaning. The complete record of how the results were obtained is a separate document in the archived artifact, the postmortem. It retells the work in the order it happened, with every result that did not survive and the full log of the registered law campaign. Pointers such as “postmortem, S16.10” refer to its sections.

## S1. THE SETUP IN FULL

The paper’s results depend on the football feed only through message size and rate, and none depends on synchronizing two clocks unless it says so.

### S1.1. Testbeds and clocks

Each of three testbeds carries a different part of the evidence (Table S3). On the cloud testbed the producer and the consumer run as separate processes on one host with 8 online CPUs, and read one wall clock, `CLOCK_REALTIME`, through `time.time_ns()`. No scheduler tunable is written anywhere in the archived campaign, which is strong evidence, short of proof, that the kernel’s defaults applied, among them the derived base slice of 3 ms (Section III-C). The clocksource was not recorded; S3 bounds it from a measured read cost. Across hosts, `chrony` reports an inter-host offset of 0.067 ms but bounds its own error at 4–7 ms per host over the 4 hosts captured, so we claim no cross-host bound.

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TABLE S1

EACH SECTION OF THE PAPER HAS ITS EVIDENCE IN NAMED SECTIONS HERE. PAPER SECTIONS BY NUMBER, THE SECTIONS OF THIS DOCUMENT THAT HOLD THEIR EVIDENCE, AND WHAT THOSE SECTIONS ADD.

| Paper section                        | Here           | What it adds                                                                                                                                                                 |
|--------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I. Introduction                      | S2, S3, S4     | The sign check’s rejections, and both failures in full.                                                                                                                      |
| II. How a benchmark times a message  | S1, S2, S5     | Testbeds, workload, settings, campaigns and where each timestamp is taken; the threshold sweep and what the check selects; origin timestamps written late in other runtimes. |
| III. Failure 1                       | S2, S3, S6, S8 | The rejection counts; every manipulation, trace and rival explanation, the exposure and the repair; the registered law campaign; the first result the check rejected.        |
| IV. Failure 2                        | S4             | The source lines, the grid tests, the registered payload manipulation, the sign channel and our own harness.                                                                 |
| V. How widespread                    | S5, S4         | The source line behind each audited tool; the registered audit of published reports; the millisecond record timestamp on Kafka’s wire.                                       |
| VI. What benchmark authors should do | S7             | The broker comparison over the runs the check keeps.                                                                                                                         |
| VII. Related work                    | S9             | The literature in full.                                                                                                                                                      |
| VIII. Threats and limitations        | S3, S4, S6, S9 | The rival explanations and the clocksource bound; the benchmark’s distributed mode; the law campaign’s verdicts; why a skew threshold does not transfer to a loaded machine. |
| IX. Conclusion                       | none           | Nothing; the conclusion draws on the sections above.                                                                                                                         |

TABLE S2

EACH TERM HAS ONE MEANING, HERE AND IN THE PAPER. THE TIMESTAMPS ARE THOSE OF EQUATIONS 1 AND 2; BOTH PROCESSES READ ONE CLOCK UNLESS A RESULT SAYS OTHERWISE.

| Term                                 | Meaning                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Failure 1                            | A latency timed from the acknowledgment turns negative when the thread recording the acknowledgment is scheduled late by more than the delivery time (Section III).                                                   |
| Failure 2                            | A millisecond timestamp and a positivity filter delete, without counting them, the sub-millisecond samples that cross no millisecond boundary (Section IV).                                                           |
| condition                            | One point of the experimental design: a broker, a send rate, a message size, a background load and a consumer pattern.                                                                                                |
| cell                                 | A condition together with a workload; where the workload is fixed, the two coincide. A run executes a cell once, and a replicate is a further run of it.                                                              |
| configuration                        | One setting of a manipulation that compares settings differing in one variable.                                                                                                                                       |
| utilization $\rho$                   | The fraction of processor time busy on the host, measured over the run.                                                                                                                                               |
| core count $k$                       | The number of cores the background load occupies.                                                                                                                                                                     |
| load geometry                        | How a utilization is reached: the load concentrated on few cores or spread over many.                                                                                                                                 |
| occupancy $p$                        | The probability that a timestamping thread is preempted and off the processor.                                                                                                                                        |
| scheduling delay                     | The time a thread that is ready to run spends queued before a core runs it.                                                                                                                                           |
| delivery time $D$                    | $t_{\text{recv}} - t_{\text{send}}$ : from the send call to the consumer holding the record.                                                                                                                          |
| acknowledgment lag $A$               | $t_{\text{ack}} - t_{\text{send}}$ : from the send call to the producer observing the broker's acknowledgment.                                                                                                        |
| transport proxy $S$                  | $t_{\text{recv}} - t_{\text{ack}} = D - A$ : a message timed from the acknowledgment. It is not a causal chain.                                                                                                       |
| end-to-end latency $E$               | $t_{\text{out}} - t_{\text{sched}}$ : from the event falling due to the consumer having handled it, the sum of the send lag, $A$ , $S$ and the handling span.                                                         |
| handling span                        | $t_{\text{out}} - t_{\text{recv}}$ : the consumer's own work on the record after $t_{\text{recv}}$ .                                                                                                                  |
| send lag                             | $t_{\text{send}} - t_{\text{sched}}$ : how late the send call ran after the event was due.                                                                                                                            |
| true delivery time $T_{\text{true}}$ | The interval $D$ would report if its timestamps were arbitrarily fine. A difference of two timestamps below zero is a negative span.                                                                                  |
| timestamp resolution $\tau$          | The smallest difference a timestamp can express: one millisecond in the OpenMessaging Benchmark.                                                                                                                      |
| grid                                 | The instants, $\tau$ apart, at which a timestamp changes value.                                                                                                                                                       |
| retention                            | The fraction of the samples a tool takes that survive into what it reports.                                                                                                                                           |
| positivity filter                    | The OpenMessaging Benchmark's rule, at <code>WorkerStats.java:95</code> , that admits a sample to its histogram only when its end-to-end latency is strictly positive.                                                |
| sign check                           | A run is rejected if more than 1% of its events carry a negative value in any latency component, or if the median of any component is negative; a condition is usable only if all of its runs survive (Section II-D). |

TABLE S3

EACH TESTBED CARRIES A DIFFERENT PART OF THE EVIDENCE. MACHINES, SOFTWARE, AND WHAT RESTS ON EACH TESTBED.

| Testbed      | Machines                                                                                                                                                              | Kernel and software                                                                                                                                                  | What rests on it                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Cloud        | Four Oracle Cloud VM.Standard.E5.Flex instances on a private network: three brokers and one driver.                                                                   | Ubuntu 22.04, kernel 6.8.0-1057-oracle (EEVDF scheduler), a 1 ms tick; CPython 3.10; chrony-disciplined clocks, logged every minute.                                 | The findings of Sections III and IV, except the law campaign's. |
| Workstation  | One Windows machine running the brokers in Docker Desktop containers, inside a WSL2 virtual machine on Microsoft's hypervisor.                                        |                                                                                                                                                                      | The sign check only; no attribution rests on it (S2).           |
| Law campaign | Three Microsoft Azure pairs in one region, each a driver with eight virtual CPUs and a broker with two: two x86 pairs (AMD EPYC 9V74) and one Arm pair (Neoverse N2). | The provider's stock kernels, 6.8.0-1064 and 6.8.0-1065; for the tick block, three kernels built from one source tree that differ in the tick alone, 1, 4 and 10 ms. | The registered law campaign (S6).                               |

### S1.2. Workload

Three properties of the football feed drive the experimental choices: it is sparse, it is bursty with one burst at a known instant, and its concurrency is bounded. The feed is StatsBomb's open event data [S1]. Our characterization covers 3,315 of its matches, and the campaigns replay eleven of them in order at a chosen send rate. A match produces a median of 3,507 events over about 8,412 s, a mean of 0.415 events per second; a match's busiest ten-second window holds 2.70 events per second at the median. Each of the eleven replayed plans opens with at least five events at one instant: two lineup records, two half-start markers and the kick-off pass. Across all 3,315 matches, 575 open with four. That burst meets the producer when it is coldest, which turns a one-off client start-up cost into the median of a short run (S8).

TABLE S4

**THREE OF THESE SETTINGS WERE LEARNED FROM THE DEFECTS THEY CAUSED. EVERY CLIENT SETTING THAT CAN MOVE A LATENCY NUMBER, AND WHY IT HOLDS ITS VALUE. LEARNED: FOUND THROUGH A DEFECT IT CAUSED, NOT CHOSEN IN ADVANCE.**

| Setting               | Value    | Why                                                                                                                                                                  |
|-----------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Kafka producer</i> |          |                                                                                                                                                                      |
| acks                  | all      | Durability matched to Redis’s synchronous XADD; a weaker acknowledgment would compare different guarantees.                                                          |
| linger.ms             | 0        | No artificial batching delay: on a sparse workload any linger would dominate the quantity measured.                                                                  |
| max-inflight          | 64       | Learned. At our producer’s default of 1 the client blocks on a round trip per event, which makes the comparison one of harnesses (S8).                               |
| compression           | none     | Football events are small JSON; compression would add CPU work to the timed path for no representative benefit.                                                      |
| ack-stamp             | callback | The default, deliberately asymmetric with Redis; the inline mode measures what the asymmetry costs (S7).                                                             |
| <i>Redis producer</i> |          |                                                                                                                                                                      |
| send workers          | pool     | Non-blocking dispatch keeps the emission loop on schedule; send and acknowledgment are timestamped in the worker.                                                    |
| <i>Consumers</i>      |          |                                                                                                                                                                      |
| Redis ack-batch       | 200      | Learned. Acknowledging each message adds a round trip per message: at 20 ms of delay, median end-to-end latency 4,138 ms against 103 ms batched (S7).                |
| Redis count           | 200      | Read batch, matched to the acknowledgment batch so neither dominates.                                                                                                |
| Kafka poll-timeout    | 1000 ms  | The client already amortizes fetches; no per-record round trip exists to remove.                                                                                     |
| <i>Replay</i>         |          |                                                                                                                                                                      |
| speed-up              | derived  | Learned. Plans carry a built-in 120× compression, so the flag is not the rate: it is derived from the plan and checked against wall time before every campaign (S8). |
| window                | 60–600 s | Swept on purpose: the number of events a run contains is itself a variable (S8).                                                                                     |

Kick-off times are recorded, so concurrency is observed: across 2,346 kick-off instants at most 12 matches kick off together, and at most 21 are in play at once. E1’s levels,  $N \in \{1, 9, 10, 12\}$  feeds, are the median kick-off slot, which is also its upper quartile, the final matchdays of an 18-team and a 20-team league, and the observed maximum; the powered campaigns kept 1, 9 and 12 (S7). The law campaign replays a steady 50 messages a second instead, except M0, which replays a match’s bursts.

### S1.3. Every setting that can move a latency number

Two clients are comparable only when matched on what they do, because Kafka and Redis expose different settings and matching them by name builds in an asymmetry. Published Kafka comparisons are frequently not comparable for this reason: client configuration and acknowledgment semantics differ silently between the systems compared [S2]. Kafka’s producer buffers and pipelines by default, while our Redis producer dispatches to a worker pool. At one request in flight, the Kafka client would wait a broker round trip per event and the Redis client would not, a difference in the harness that a reader would attribute to the broker. Table S4 states every client setting that can move a latency number, and why it holds its value.

### S1.4. Which campaign settles which claim

Each campaign manipulates one variable and holds the others fixed, so each settles one claim and no claim rests on a single cell. Table S5 maps each claim to the campaign behind it, the runs per cell and the section here that reports it. H1, H2 and H3 label hypotheses the campaigns tested, and E1 is the historical broker comparison on the cloud testbed. The injected delay cancels in the transport proxy, so the delay sweep is a negative result only (S8).

### S1.5. Where each timestamp is taken, and what each span covers

Of the spans the harness reports, only the transport proxy lacks a causal order between its two timestamps, and only it inverts (Table I). Figure S1 draws the four timestamps of Equation 1 and the fifth,  $t_{\text{out}}$ , at which Equation 2 closes the end-to-end latency. The send lag,  $A$  and  $D$  each start from a timestamp taken before the event that ends them can happen, so all three are causal chains and a late thread can only lengthen them.  $S$  subtracts two branches of the broker’s append, with no order between them, so it is negative whenever  $A > D$  (Equation 3).

The two clients read the acknowledgment timestamp in different places, but in both it is read only after a thread that was not running has been made runnable and then scheduled. The send timestamp is read on the calling thread immediately before the client is invoked. Under Redis the acknowledgment timestamp is read on that same thread the moment the synchronous `xadd` returns, with no second thread involved. Under Kafka it is read, by default, in the client’s I/O thread when the delivery

TABLE S5

**NO CLAIM RESTS ON A SINGLE CELL.** EACH CLAIM, THE CAMPAIGN BEHIND IT, THE RUNS PER CELL, AND THE SECTION HERE THAT REPORTS IT. WHERE THE SIGN CHECK REMOVES RUNS, S2 BOUNDS THE SELECTION THIS INTRODUCES.

| Claim                                 | Campaign                    | Runs per cell        | Here   |
|---------------------------------------|-----------------------------|----------------------|--------|
| Sign-check rejections                 | both testbeds               | 2,266 runs in total  | S2     |
| Rate against utilization (H2)         | E-A, E-A3                   | 25 per load level    | S3     |
| Mixture structure                     | E-A3, nine levels           | 25 per level         | S3     |
| Priority at fixed utilization         | E-A5, E-A5b, E-A7           | 25 per configuration | S3     |
| Load geometry                         | E-A6, both geometries       | 25 per configuration | S3     |
| Traced run-queue stalls               | E-A9                        | 25 per configuration | S3     |
| $T_{\text{true}}$ sweep               | E-A10, four payloads        | 25 per level         | S3     |
| Acknowledgment timestamping (H3)      | E-C3, replicated in E-C4    | 10, then 30 per cell | S3, S7 |
| Retention under the positivity filter | the benchmark, instrumented | 223 runs in total    | S4     |
| Broker comparison                     | powered, replicated         | 15, then 8 per cell  | S7     |
| Injected delay (H1)                   | co-located against network  | 15 per delay level   | S8     |
| Per-run start-up cost                 | window sweep, twice         | 3 per window         | S8     |
| Historical comparison                 | E1                          | 8–36, 52.8–100% kept | S7, S8 |

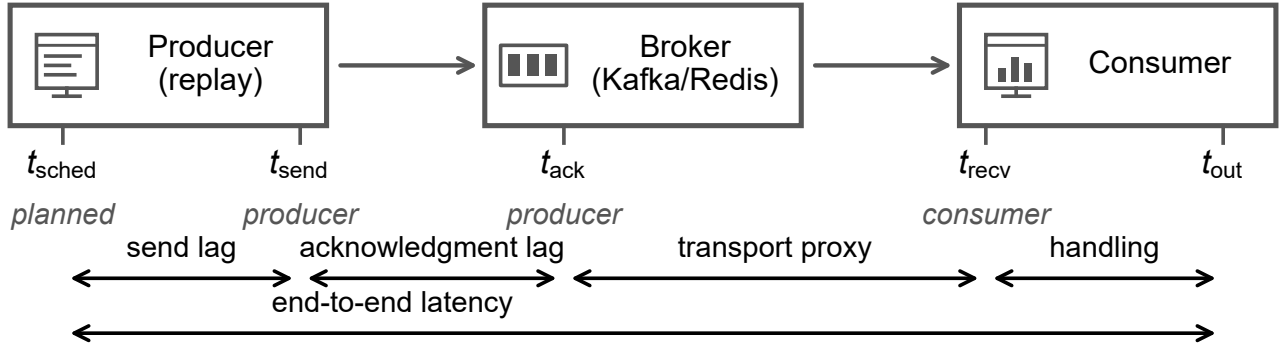


Fig. S1. **Five timestamps bracket a message's path.** A replay process paces recorded match events at a chosen rate into a broker, Kafka or Redis Streams, and a consumer reads them back. The upper four spans are the terms of Equation 2 and sum to the fifth, the end-to-end latency  $E$ .

callback fires. Our Kafka producer's inline mode instead timestamps on the calling thread once the send future resolves, and requires one request in flight.

The consumer reads  $t_{\text{recv}}$  when it holds the record, and an output timestamp  $t_{\text{out}}$  once the record is in hand. Our Kafka consumer parses the payload inside `poll()`, before  $t_{\text{recv}}$ ; our Redis consumer parses it between the two. The handling span  $t_{\text{out}} - t_{\text{recv}}$  is therefore 281 ns under Kafka and 19,480 ns under Redis, medians of run medians over 5,913 runs, and negative on none of the 738,730 events. The parse thus falls inside  $D$  for Kafka only, and it is worth 2.4% of Redis's median delivery time of 800  $\mu\text{s}$ . A longer  $D$  produces fewer negative spans, and Kafka already has the higher negative-span rate, so correcting the asymmetry would widen the gap between the brokers in Table I. The end-to-end latency runs to  $t_{\text{out}}$ , not  $t_{\text{recv}}$ , so it contains both parses and is unaffected.

## S2. THE SIGN CHECK, AND WHAT IT SELECTS

Where the one-per-cent line sits changes how many runs the sign check rejects, and no conclusion the paper draws. The rule is that of Section II-D. A run is rejected if more than 1% of its events carry a negative value in any latency component, or if the median of any component is negative; a condition is usable only if all of its runs survive. We fixed the rule before the final campaign but after earlier campaigns had run, which is weaker than pre-registration, and the threshold sweep below exists partly to compensate. On each testbed the rule rejected more than half of the runs (Table S6).

### S2.1. The threshold sweep

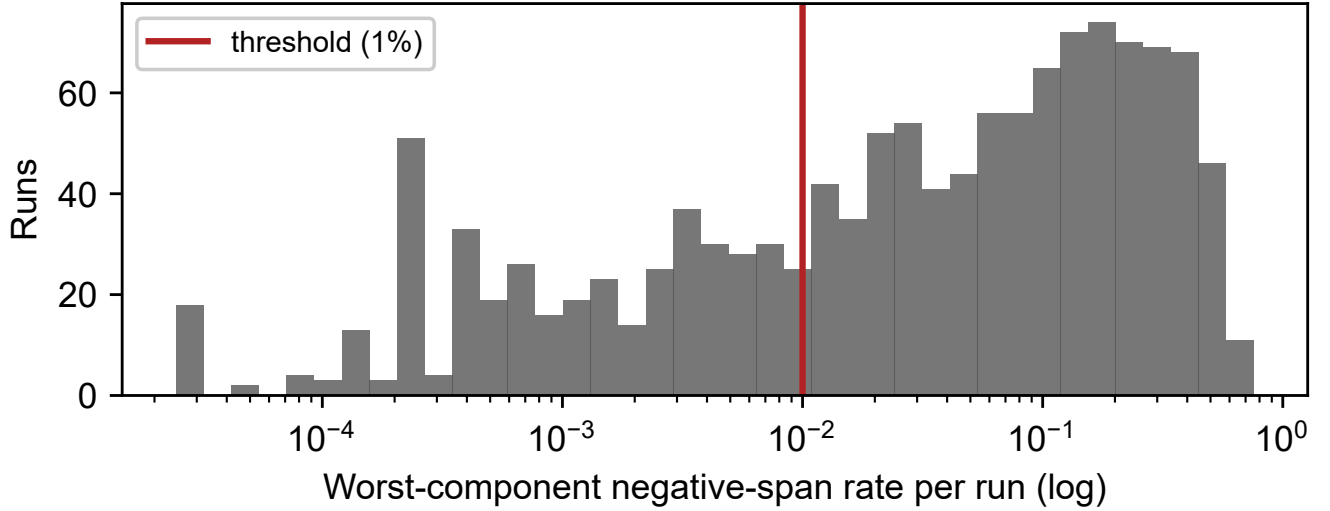
Per-run rates offer no natural threshold, so the line had to be chosen. On the workstation testbed 104 runs carry no negative span at all, and the rest spread over more than four decades of rate (Fig. S2a). Raising the threshold lowers the share of runs rejected (Fig. S2b), and at one per cent the rule rejects 62.4% of the 1,382. The verdict on our first result does not move with it. Swept from 0 to 20%, the threshold makes none of that result's cells usable, and the best of them keeps 23 of its 30 runs (S8).

TABLE S6

**MORE THAN HALF OF THE RUNS ON EACH TESTBED FAIL THE SIGN CHECK. RUNS BEHIND A REPORTED RESULT, BY TESTBED; A CONDITION IS USABLE ONLY IF ALL OF ITS RUNS SURVIVE. THIS CORPUS DIFFERS FROM THE SPAN RECOUNT OF TABLE I, WHICH COVERS EVERY CLOUD RUN THAT PRODUCED MATCHED EVENTS.**

| Testbed     | Runs  | Rejected      | Conditions | Usable |
|-------------|-------|---------------|------------|--------|
| Workstation | 1,382 | 862 (62.4%)   | 76         | 8      |
| Cloud       | 884   | 459 (51.9%)   | 40         | 13     |
| Both        | 2,266 | 1,321 (58.3%) | 116        | 21     |

(a) 104 of 1,382 runs with no negative span



(b) The threshold is a real choice

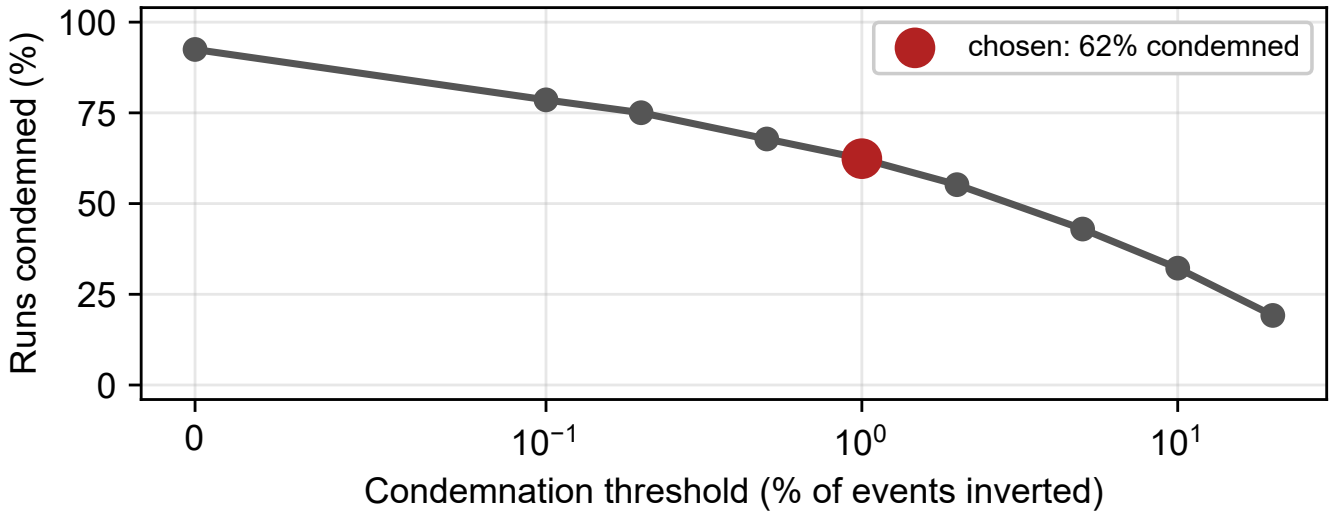


Fig. S2. **No natural threshold, and the choice moves no conclusion.** The sign check on the workstation testbed. (a) Per-run negative-span rates in the worst latency component: 104 runs carry none at all, and the rest spread over more than four decades. The one-per-cent rule passes more runs than those 104, and rejects 62.4% of the 1,382. (b) Share of runs rejected against the threshold, swept from 0 to 20% of events.

### S2.2. What the check selects, bounded

The check rejects unequally between configurations, so the corpus it leaves is not a random sample of the corpus measured. This is the most serious internal threat to any comparison drawn from the survivors, and we bound it in the way each corpus allows (Table S7). E1, the historical broker comparison on the cloud testbed, lost its rejected values. Panel (a) therefore imputes every rejected Redis run at the largest value observed in its own cell, and the worst-case Hodges–Lehmann shift stays inside

TABLE S7  
SELECTION BY THE SIGN CHECK DOES NOT DECIDE THE BROKER COMPARISON. (A) E1, WHOSE REJECTED RUNS' VALUES ARE LOST. (B) THE POWERED TRANSPORT CAMPAIGN AND ITS REPLICATION, WHOSE REJECTED RUNS' VALUES SURVIVE.

| (a) E1 |            |            |                 |               |
|--------|------------|------------|-----------------|---------------|
| $N$    | Redis kept | Shift (ms) | Worst case (ms) | Inside margin |
| 1      | 8/8        | +0.081     | +0.081          | yes           |
| 9      | 20/27      | +0.021     | -0.051          | yes           |
| 10     | 17/30      | +0.116     | -0.485          | yes           |
| 12     | 19/36      | +0.053     | -0.227          | yes           |

Worst case: every rejected Redis run imputed at the largest value observed in its cell. Margin 1 ms.

| (b) Powered transport campaigns |     |                |          |           |         |            |
|---------------------------------|-----|----------------|----------|-----------|---------|------------|
| Campaign                        | $N$ | Redis kept (%) | All runs | Runs kept | Change  | $V^*$ (ms) |
| powered                         | 1   | 53.3           | 0.4095   | 0.4084    | -0.0011 | none       |
| powered                         | 9   | 43.7           | 0.4181   | 0.4156    | -0.0025 | 0.60       |
| powered                         | 12  | 37.0           | 0.4197   | 0.4170    | -0.0027 | 0.60       |
| replication                     | 1   | 25.0           | 0.3969   | 0.3807    | -0.0162 | 0.55       |
| replication                     | 9   | 45.8           | 0.4120   | 0.4137    | +0.0017 | 0.60       |
| replication                     | 12  | 39.8           | 0.4132   | 0.4157    | +0.0025 | 0.55       |

All runs, runs kept: the Hodges–Lehmann shift, Kafka minus Redis, in ms. Change: kept minus all.  $V^*$ : the smallest value which, imputed to every rejected Redis run, would flip the shift's sign. The rejected runs' own medians center on 0.10–0.12 ms.

the 1 ms margin in every cell. That bound holds only while retention exceeds one half, and the tightest cell sits 2.8 points above that line.

The powered transport campaigns, on which the broker equivalence of Section VI rests, fall below that line. The check keeps 25.0–53.3% of their Redis runs, under one half in five of the six cells. Their rejected runs' values survive, so panel (b) computes the selection directly. Restoring those runs moves the Hodges–Lehmann shift by at most 0.016 ms. Reversing its sign would take every one of them at 0.55–0.60 ms, about five times the 0.10–0.12 ms at which their own medians center. In the sixth cell no value reverses it.

A run that passes has cleared the cheapest available bar and nothing more. Load-induced inflation that stays positive passes the check by construction (Section III-A).

### S3. FAILURE 1: THE FULL EVIDENCE

Failure 1 comes from the scheduler. Every manipulation of the timestamping thread's wait for a core, or of the delivery time that wait must outlast, moved the negative-span rate the way the mechanism predicts. A kernel trace that times the wait predicts the rate to within a third, unfitted. The evidence follows the order of Section III: three manipulations, the model they support, the trace, the rivals, the timescale and the cost.

#### S3.1. Real-time priority cuts the rate in every matched pair

Moving the timestamping threads to real-time priority cut the negative-span rate between 7- and 80-fold in each of eight matched pairs, with utilization unchanged. Under `SCHED_FIFO` the threads preempt the background load instead of queueing behind it, so their occupancy falls while the machine stays as busy. An occupancy mechanism predicts a collapse toward the running-state floor. Any account in which utilization alone sets the rate predicts no change, and the M/G/1 waiting-time form we had adopted and registered is such an account (S8).

The pairs come from three campaigns (Table S8). E-A5 is the original manipulation, E-A5b widened the load ladder to 60 and 95%, and E-A7 replicated E-A5. A rule fixed in advance withholds a pair whose two achieved utilizations differ by more than 0.05. It withheld 0 pairs, and the widest disagreement in the table is 0.002, 25 times inside the tolerance.

Table S8 shows which configuration moves. The real-time configuration stays between 0.0017 and 0.0144 over the whole ladder, while the ordinary one climbs with load from 0.046 to 0.30. That is Equation S2 with  $p$  moved directly. The factor grows with load as a trend and not as a law, since 60% gives  $27\times$  and 70% gives  $7\times$ , so the factor is reported as a range. The real-time rates settle on both sides of the floor the model names in advance,  $C_0 \approx 0.004$ , which is therefore a scale and no bound.

Table II prints each factor without its interval. For the priority pairs the Katz 95% intervals are [21–74] at 75% load ( $z = 19.8$ ) and [33–86] at 88% ( $z = 31.9$ ). For the load geometry they are [1.80–2.39] in the original campaign ( $z = 10.3$ ) and [1.73–2.43] in the replication ( $z = 8.4$ ). The priority intervals are wide because the collapsed configurations are sparse. The Wilson intervals on the rates themselves are disjoint in every pair.

TABLE S8

**FACTOR OF 7 TO 80, WITH UTILIZATION MATCHED IN EVERY PAIR.** ORDINARY AND REAL-TIME NEGATIVE-SPAN RATES AT THE SAME ACHIEVED UTILIZATION, AND THE FACTOR BETWEEN THEM, FROM THREE CAMPAIGNS. BOTH CONFIGURATIONS' ACHIEVED  $\rho$  IS PRINTED SO THAT THE MANIPULATION CAN BE CHECKED PAIR BY PAIR. GENERATED AT BUILD TIME FROM THE CAMPAIGN FILES.

| campaign | load | $\rho$ ordinary | $\rho$ real-time | ordinary | real-time | factor |
|----------|------|-----------------|------------------|----------|-----------|--------|
| E-A5b    | 60%  | 0.6055          | 0.6035           | 0.0459   | 0.0017    | 27×    |
| E-A5b    | 70%  | 0.7032          | 0.7025           | 0.1032   | 0.0144    | 7×     |
| E-A5     | 75%  | 0.7531          | 0.7525           | 0.1320   | 0.0034    | 39×    |
| E-A7     | 75%  | 0.7550          | 0.7537           | 0.1504   | 0.0044    | 35×    |
| E-A5b    | 88%  | 0.8806          | 0.8809           | 0.2214   | 0.0034    | 66×    |
| E-A5     | 88%  | 0.8809          | 0.8812           | 0.3049   | 0.0057    | 54×    |
| E-A7     | 88%  | 0.8825          | 0.8825           | 0.2526   | 0.0054    | 47×    |
| E-A5b    | 95%  | 0.9501          | 0.9501           | 0.2938   | 0.0037    | 80×    |

TABLE S9

**AT  $k = 6$  ONE UTILIZATION GIVES TWO RATES, IN TWO CAMPAIGNS.** CONCENTRATED LOAD LEAVES  $C - k$  CORES FREE; SPREAD LOAD DUTY-CYCLES ALL OF THEM AND LEAVES NONE. EVERY CELL IS 2,985 MATCHED EVENTS OVER 25 RUNS, AND  $z$  IS A TWO-PROPORTION TEST BETWEEN THE GEOMETRIES OF A ROW. ACHIEVED  $\rho$  AT  $k = 6$  IS 0.7531 IN ALL FOUR CONFIGURATIONS; AT  $k = 5$  IT IS 0.6284/0.6250 AND 0.6284/0.6259, AND AT  $k = 7$  0.8778/0.8809 AND 0.8778/0.8822.

| $k/C$ | E-A6 (original) |        |       | E-A6b (replication) |        |      | factors      |
|-------|-----------------|--------|-------|---------------------|--------|------|--------------|
|       | concentrated    | spread | $z$   | concentrated        | spread | $z$  |              |
| 5/8   | 0.0201          | 0.0379 | 4.09  | 0.0111              | 0.0509 | 8.89 | 1.88×, 4.61× |
| 6/8   | 0.0824          | 0.1709 | 10.27 | 0.0600              | 0.1229 | 8.44 | 2.07×, 2.05× |
| 7/8   | 0.2281          | 0.2415 | 1.22  | 0.1973              | 0.2342 | 3.46 | 1.06×, 1.19× |

### S3.2. One utilization, reached two ways, gives two rates

Utilization is not the variable, because one utilization reached two ways gives two rates. A load that runs  $k$  of the  $C = 8$  cores flat out leaves the other  $C - k$  free, while one that duty-cycles every core leaves none. If the timestamping thread fails when it cannot find a free core, the spread load should be worse at equal utilization, and it is (Table S9).

At  $k = 6$  both geometries sit at  $\rho = 0.7531$ , identical to four decimals, on one machine in one hour. Their rates differ 2.07-fold ( $z = 10.3$  over 2,985 matched events per cell), and a function of  $\rho$  returns one value for one input. The replication E-A6b reproduces that cell almost exactly, at the same  $\rho$  and a factor of 2.05. It also reproduces the direction at  $k = 5$ . At  $k = 7$  the difference is small, since one free core in eight is nearly as useless as none, and only the replication resolves it (1.19×,  $z = 3.46$ ). Multi-server queueing expects geometry to matter at fixed  $\rho$ . What the contrast refutes is the single-parameter form, in which  $\rho$  alone fixes the rate.

### S3.3. A longer path inverts less often

A slower path is a more reliable measurement, because the same stalls must outlast a longer interval to invert it. Padding the payload lengthens the path without touching the scheduler. In E-A10 the median of the transport proxy  $S$  rose 76.9-fold over 4 levels while the negative-span rate fell 4.1-fold [Katz 95%: 3.5–4.8], monotonically, and a repeat on another day agrees (Table S10). Serialization pushes utilization slightly up, against the observed fall, so the effect is measured against its own confound. A stress account predicts the opposite direction.

The fall is shallow: a 77-fold longer threshold bought only a 4.1-fold lower rate. Least squares of the log rate on the log median of  $S$ , which stands in for  $T_{\text{true}}$ , gives

$$\Pr[S < 0] = 0.238 T_{\text{true}}^{-0.339}, \quad R^2 = 0.990, \quad (\text{S1})$$

TABLE S10

**SLOWER PATH, MORE RELIABLE MEASUREMENT, IN TWO CAMPAIGNS.** PAYLOAD PADDING IN TWO CAMPAIGNS. THE MEDIAN OF  $S$  SPANS 76.9× AND 76.8× WHILE THE RATE FALLS 4.09× AND 4.26× [3.62–5.01], BOTH MONOTONICALLY, WITH UTILIZATION HELD TO THE  $\rho$  SPAN SHOWN. EACH CELL IS 2,985 EVENTS.

| Padding (B) | E-A10           |        | E-A10b          |        | $\rho$ span |
|-------------|-----------------|--------|-----------------|--------|-------------|
|             | Median $S$ (ms) | Rate   | Median $S$ (ms) | Rate   |             |
| 0           | 0.701           | 0.2603 | 0.761           | 0.2338 | 0.881–0.882 |
| 4,096       | 2.381           | 0.1906 | 2.474           | 0.1652 | 0.882–0.883 |
| 65,536      | 14.638          | 0.0888 | 16.059          | 0.0784 | 0.885–0.885 |
| 262,144     | 53.898          | 0.0637 | 58.482          | 0.0549 | 0.893–0.895 |

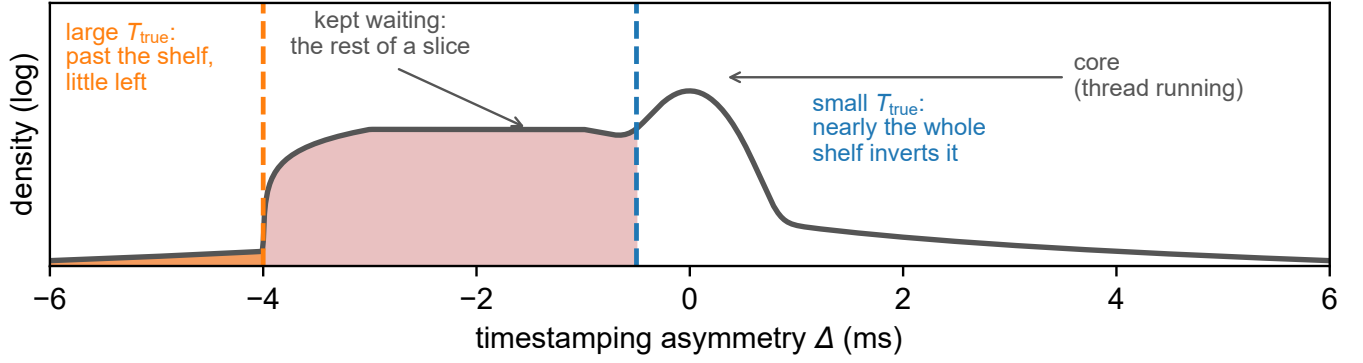


Fig. S3. In the model, a short delivery time is inverted by nearly the whole waiting state, and one longer than a slice and a tick by none of it. The two-state model, drawn. The timestamping asymmetry  $\Delta$  has a narrow core while the timestamping thread runs. While it waits, a shelf reaches from the core to  $-3$  ms, the scheduler’s base slice, and a tick beyond. The shape is schematic: two components drawn with the derived slice and tick, not fitted. The shelf, a wait for the rest of a slice, was read from the law campaign’s recordings after its verdicts (S6.5). The measured distributions are the stall spectrum of Figure 3 and the per-acknowledgment waits of S6.5.

TABLE S11

FROM IDLE TO THE KNEE THE RATE GROWS 61-FOLD AND THE JITTER ONLY 5-FOLD. E-A3, NINE LOAD LEVELS, 25 RUNS EACH, PRE-REGISTERED. THE JITTER WIDTH IS  $\sigma_{\text{core}} = \text{IQR}/1.349$ . NEGATIVE SPANS CLUSTER AT EVERY LOAD (RUNS-TEST  $z$ ). THE THREE  $\rho = 1$  ROWS ARE DISTINCT OVERSUBSCRIPTION LEVELS THAT THE UTILIZATION COORDINATE CANNOT SEPARATE.

| Utilization $\rho$ | Jitter width $\sigma_{\text{core}}$ (ms) | Negative-span rate | Clustering $z$ |
|--------------------|------------------------------------------|--------------------|----------------|
| 0.003              | 0.126                                    | 0.004              | −6.8           |
| 0.253              | 0.138                                    | 0.004              | −6.8           |
| 0.504              | 0.180                                    | 0.006              | −6.8           |
| 0.628              | 0.181                                    | 0.024              | −6.9           |
| 0.753              | 0.238                                    | 0.076              | −5.7           |
| 0.878              | 0.631                                    | 0.224              | −4.9           |
| 1.000              | 1.224                                    | 0.371              | −4.3           |
| 1.000              | 1.293                                    | 0.317              | −4.7           |
| 1.000              | 1.710                                    | 0.261              | −4.4           |

with a Student- $t$  95% interval on the exponent of 0.234–0.443 at 2 degrees of freedom. E-A10b returns 0.344 against 0.339, while the prefactor drifts (0.238 against 0.216). The four points of Table S10 describe the sweep and license no statement about the moments of the stall distribution (postmortem, S8). What they do show is the delivery factor  $G(T_{\text{true}})$  of Equation S2 as an experiment: the same stall distribution overlaps a short delivery time almost entirely and a long one hardly at all, where a load-driven account predicts no change.

#### S3.4. Two states account for the rate, and the second is the run queue

Two states are the smallest model consistent with these data. A timestamping thread is either running, and timestamps late by a narrow jitter, or waiting off the processor. Condition the identity of Equation 3 on the delivery time. Write  $p$  for the thread’s occupancy of the waiting state and  $G$  for the survival function of its residual delay there. To leading order,

$$\Pr[S < 0 \mid T_{\text{true}}] = p G(T_{\text{true}}). \quad (\text{S2})$$

Load moves  $p$  without fixing it (S3.2), so the form is written without a load term. Figure S3 draws it.

Clustering follows by construction. Being preempted is an interval, so the events inside it are corrupted together. A Wald–Wolfowitz runs test on each run’s sequence of span signs finds strong clustering at every load of E-A3, even with no background load (Table S11). Coarse kernel timestamps would corrupt events independently and give  $z \approx 0$ . From idle to the knee of the load ladder, E-A3’s negative-span rate grows 61-fold while the jitter width grows only 5-fold. Two parameters move at very different rates: the jitter width  $\sigma_c$  belongs to the running state and  $p$  to the scheduler. A scale family has one parameter, which is why it fails here. The highest rate, 0.37 at  $\rho = 1$ , is the largest over the conditions measured and not a bound, as the floor  $C_0$  of S3.1 is a scale and not a bound.

The model also predicts a shape. A scale family changes  $\sigma$ , so its tail curves rescale horizontally. The two-state model changes  $p$ , so its curves shift vertically and stay parallel on a log scale, and the log ratio of two conditions’ tail masses should not depend on the threshold. Over tail estimates backed by at least 20 events, that log ratio’s spread over thresholds has a median of 0.29 across conditions, a factor of 1.34, and a worst case of 0.91. The scale-family collapse fails by  $23\times$ .

TABLE S12

**KERNEL TRACING PREDICTS THE NEGATIVE-SPAN RATE, UNFITTED, IN EVERY CONFIGURATION THAT HAS ONE.**  $\Pr[\text{STALL} > 0.5 \text{ ms}]$  FROM THE `SCHED_WAKEUP` AND `SCHED_SWITCH` TRACEPOINTS, AGAINST THE NEGATIVE-SPAN RATE FROM APPLICATION TIMESTAMPS IN THE SAME CELL. EVERY CELL IS 2,985 EVENTS. EVERY TRACED REAL-TIME CONFIGURATION READS EXACTLY ZERO WHERE THE UNTRACED CONTROL READS 15 IN 2,985, SO THAT ZERO IS THE TRACER'S.

| Campaign                                                     | Configuration | Load | $\rho$ | Traced wakeups | $\Pr[\text{stall} > 0.5 \text{ ms}] / \text{rate}$ | Ratio             |
|--------------------------------------------------------------|---------------|------|--------|----------------|----------------------------------------------------|-------------------|
| E-A9                                                         | ordinary      | 88%  | 0.8822 | 551,956        | 0.181 / 0.231                                      | 0.78              |
| E-A9                                                         | real-time     | 88%  | 0.8819 | 570,591        | 0.018 / 0.000                                      | —                 |
| E-A9b                                                        | ordinary      | 75%  | 0.7543 | 709,819        | 0.168 / 0.128                                      | 1.32              |
| E-A9b                                                        | real-time     | 75%  | 0.7531 | 641,308        | 0.016 / 0.000                                      | —                 |
| E-A9b                                                        | ordinary      | 88%  | 0.8825 | 687,986        | 0.207 / 0.196                                      | 1.06 <sup>†</sup> |
| E-A9b                                                        | real-time     | 88%  | 0.8822 | 649,788        | 0.016 / 0.000                                      | —                 |
| Untraced control: E-A9's first attempt, probe never attached |               |      |        |                |                                                    |                   |
| —                                                            | ordinary      | 88%  | 0.8812 | —              | — / 0.272                                          | —                 |
| —                                                            | real-time     | 88%  | 0.8809 | —              | — / 0.005                                          | —                 |

<sup>†</sup>Withheld by the tracer check, 28% drift against a 25% rule fixed in advance: shown, not used.

Separability therefore holds to first order, with the two largest spreads at intermediate load. The test was exploratory, and a confirmatory version was registered and run, the decisive campaign being the priority manipulation of S3.1.

On the load axis alone the model has no content, since with  $p$  unconstrained any monotone curve in  $\rho$  can be written  $p(\rho)G$ . A fit on that axis cannot credit the jitter width either. Restoring it to the form reaches  $R^2 = 0.9905$ , and freezing it at its mean does better, 0.9982, because the jitter width and  $\rho$  move together on this ladder. The model's content is on the delivery-time axis, where S3.3 tests it.

The campaigns measure  $p$  only as a time average. E-A7 registered that real-time priority would cut the thread's occupancy at least tenfold. Its scheduler counters show the occupancy falling 1.7- and 2.2-fold at the two loads, and the mean stall 3- to 5-fold, against a 35- to 47-fold fall in the rate: the registered prediction failed. Counters that report means cannot see a tail event, which is why the direct trace of S3.5 was needed. The campaigns therefore confirm the mechanism and its consequences and leave the factorization untested. Those consequences stand without it. The direction under payload follows from Equation 3, the floor  $C_0$  is the  $p \rightarrow 0$  limit, and the trace of S3.5 compares two probabilities. The registered campaign later timed the waiting state for every acknowledgment and measured the shape of  $G$ , in a reading made after its verdicts (S6.5).

### S3.5. A kernel trace predicts the rate unfitted, and changes it

The negative-span rate matches, to within a third, the probability that a run-queue stall outlasts the delivery time, measured without reference to any negative span. With `bpfftrace` on the `sched_wakeup` and `sched_switch` tracepoints we recorded the delay between a timestamping thread becoming runnable and reaching a core, for every such transition in a run. The two quantities share no probe, no estimator and no data. One is a count of negative spans in application records, the other a kernel histogram of scheduler transitions. Across the three ordinary configurations of Table S12 the ratio of traced probability to measured rate is 0.78, 1.06, 1.32, with nothing fitted, and the tracer check below withholds one of the three. The residual has no consistent sign, so no bias is resolvable inside a third.

The tracer is not a free observer, and its cost was measured. An untraced twin, a first attempt whose probe failed to attach, ran two hours apart with the same design and measured 0.272 against the traced 0.231 ( $z = 3.6$ ), 14.8% lower. That gap lies inside the drift between campaigns: the ordinary 88% configuration spans 0.221 to 0.305 over its four untraced measurements, and the replication's traced run reads 0.196. It limits the level, on which no claim rests. The agreement above compares two quantities from one run, which a common perturbation moves together, and each prediction is held against the traced run's own rate.

A rule fixed in advance decides which traced configurations are compared. One whose rate differs from its untraced comparator by more than 25% is not. The replication's 88% configuration measures 0.1956 against the control's 0.272, a drift of 28%, so its comparison is withheld. Its row stays in the table with the ratio it would have given, 1.06.

The tracer also erased the smallest rate. Every traced real-time configuration, three across two campaigns and two loads, recorded no negative span in 2,985 events, where the untraced twin of one of them recorded 15. At that rate the chance of one traced configuration showing none is  $3 \times 10^{-7}$ , and of all three below  $10^{-17}$ . Attaching BPF to `sched_switch` therefore suppressed the negative spans in the real-time configuration. No claim rests on that configuration's level. The tracer changed the quantity it came to observe, and it did so where that quantity was smallest.

### S3.6. Ten rival explanations, each excluded or bounded

No rival to the run queue survives the corpus, and four fall to the manipulations already shown (Section VIII). One clock excludes clock skew. Identical utilization at rates twofold apart excludes utilization. `SCHED_FIFO` excludes timer noise, which

does not respect scheduling class, and a longer path that lowers the rate excludes stress. Coarse kernel timestamps fall to the runs test of S3.4. The other five need an argument each.

*The second thread:* Kafka’s client timestamps the acknowledgment on its I/O thread when the delivery callback fires, and Redis’s on the calling thread the moment `XADD` returns (Figure 2; S1.5). The harness was built expecting that hand-off to cause the negatives. The hand-off is real: timestamping Kafka on the calling thread shrinks the gap between the brokers’ medians by 0.071 ms [95% bootstrap over runs: 0.039–0.113], about a quarter, and by 0.039 ms in a replication. It is not the mechanism. Redis, which has no second thread, inverts on 8.19% of its events against Kafka’s 8.67%, and on more of its runs (88% against 65%). A thread returning from a blocking socket read must also be woken and scheduled before it reads the clock, and that wait is charged exactly as a callback’s is. The cleaner control, Kafka’s rate under both timestamping modes, cannot be computed, because that campaign kept only its medians.

*The scheduler:* The scheduler works as designed. A base slice with a wakeup-preemption check trades wakeup latency for throughput and fairness, and the escapes from it are opt-in: real-time priority, a preemption-free build, or a pinned busy-polling core. The kernel’s `SO_TIMESTAMPING` already returns the time a packet arrived, and none of the tools we audited uses it.

*The hypervisor:* The hypervisor is bounded, not excluded, because both testbeds are virtualized: the cloud instances are virtual machines, and the workstation runs its brokers in Docker Desktop containers inside a WSL2 virtual machine on Microsoft’s hypervisor. Two observations disagree with an account in which stolen processor time causes the negatives. Real-time priority is a guest-side control, which stolen virtual-CPU time would not yield to, and it cuts the rate 7- to 80-fold. The upper mode of the stall spectrum sits on the guest kernel’s own slice arithmetic (S3.7), on no quantity the hypervisor exposes. Steal time may still contribute, and no testbed here can show how much.

*The interpreter lock:* The interpreter lock is present in principle and bounded in practice. The timestamping threads are CPython, which releases the lock around blocking I/O and retakes it on return, and the clock read is a Python call. A timestamp therefore waits for a core and then for the lock. The kernel trace cannot see the second wait, which happens after the switch, in user space. Priority does not separate the two, since a `SCHED_FIFO` thread also reaches the lock sooner, and nor does magnitude, since the default switch interval and the base slice are both of millisecond order. The direction of the disagreement settles it. A second wait in series with the run queue would leave an estimator built from the run queue alone short of the observed rate, and its ratios, 0.78, 1.06, 1.32, straddle unity instead. The registered campaign found the same failure in Kafka’s own Java client, which has no interpreter lock, at a lower rate (S6.3).

*The clock:* Clock incoherence across virtual CPUs [S3] is excluded twice. First, a measurement bounds the clocksource. The platform probe’s smallest observed increment of `clock_gettime(CLOCK_MONOTONIC)` was 90 ns, an upper bound on the cost of one read. That excludes `acpi_pm` by arithmetic, since its tick is 279 ns, and `hpet` by its published read cost near 1.2  $\mu$ s. What remains, `kvm-clock` or `tsc`, fits the bound, and each carries the coherence guarantee. A `kvm-clock` that fast is on the vDSO path, which `kvmclock.c` enables only when the hypervisor sets `PVCLOCK_TSC_STABLE_BIT`, the flag that guarantees monotonic time across CPUs. A KVM guest selects `tsc` only when the kernel finds it constant, non-stop and stable, and re-rates it away if it later warps. Which clocksource ran is unrecorded, and the published read costs vary about twofold, against an order-of-magnitude margin. Second, the corpus leaves no headroom. An offset between clocks large enough to produce the deepest negative span, –99.8 ms, would have driven the send-referenced span far below zero. That span’s floor, 220  $\mu$ s, bounds any such offset 455-fold below what the deepest negative span needs.

### S3.7. The upper stall mode survives coarsening, and its bucket holds the derived slice

The upper mode of the traced stall spectrum survives a change of bucket edges, and the bucket it occupies holds the scheduler’s derived base slice. A count of local maxima belongs jointly to the counts and to the bin edges, and those edges came from `bpfttrace` as the millisecond came from the benchmark. The per-event stall durations were not retained, so coarsening is the only test available. Merging adjacent buckets doubles the bin width, and starting the merge at the first bucket or the second puts the edges in different places. The spectrum has 3 maxima on the tool’s own 16 buckets, 3 again under both phasings at 2 times the width, and fewer at 4 times, so the count holds to within a factor of 2 in bin width.

Above the upper mode the counts collapse. They fall 5.0-, 3.4- and 4.7-fold over the next three octaves and then 357-fold, where a power law would fall by one factor every time. Power laws fitted below the mode and above it both fail a bootstrap goodness-of-fit ( $p < 0.0004$  and  $p < 0.0004$ , over 2,500 replicates each).

The slice is derived from the kernel’s rule and the campaign’s own data. Each geometry condition loads  $k$  cores and records the achieved utilization, so  $k/\rho$  recovers 8 cores six times over, with a spread of 0.05. The kernel’s rule scales a 0.75 ms constant by one plus the integer base-two logarithm of the CPU count, up to eight CPUs, giving  $4 \times 0.75 \text{ ms} = 3 \text{ ms}$  [S4]. The constant is specific to this kernel and became 0.70 ms in v6.15, after our 6.8. The build sets `CONFIG_HZ = 1000`, a 1 ms tick. With run-to-parity, a woken thread that loses the wakeup-preemption check waits for the incumbent to finish its slice. The same posting notes that a task notices its deadline only at the next tick, so waits end in a band a tick wide. No scheduler tunable is written anywhere in the archived campaign, so the derived value is the one that applied. That is strong evidence, short of proof, and a future campaign should read `base_slice_ns` and `CONFIG_HZ` off the image under test. How long a woken thread actually waits was timed later by the registered campaign: what is left of a slice, up to a whole one, in a reading made after its verdicts (S6.5).

TABLE S13

**BELOW 0.72 MS THE MEDIAN ACKNOWLEDGMENT LAG EXCEEDS THE DELIVERY TIME IT CORRECTS.** COMPUTED FROM THE MEASURED ACKNOWLEDGMENT LAGS. LEFT: THE ERROR  $A/T_{\text{true}}$  ON ONE REPORTED LATENCY AT THE CORPUS'S MEDIAN LAG, 725  $\mu\text{s}$ . MIDDLE: THE SAME ERROR AT THE TENTH AND NINETIETH PERCENTILES OF THE LAG ACROSS CONDITIONS, 500 AND 1900  $\mu\text{s}$ . RIGHT: THE GAP A COMPARISON WOULD REPORT BETWEEN TWO IDENTICAL SYSTEMS WHOSE CLIENTS DIFFER ONLY IN WHERE THE ACKNOWLEDGMENT IS TIMESTAMPED, A CALLBACK THREAD AGAINST AN INLINE TIMESTAMP. WHERE IT READS "INVERTS" THE APPARENT GAP HAS CHANGED SIGN. ON EVERY ROW BELOW THOSE NOTHING IS NEGATIVE, AND THE SIGN CHECK FIRES ON NONE OF IT.

| True path<br>$T_{\text{true}}$ | Error on one<br>measurement | p10–p90<br>across conditions | Apparent gap between<br>two identical systems |
|--------------------------------|-----------------------------|------------------------------|-----------------------------------------------|
| 0.25 ms                        | 290%                        | 200–760%                     | inverts                                       |
| 0.5 ms                         | 145%                        | 100–380%                     | inverts                                       |
| 1 ms                           | 72%                         | 50–190%                      | inverts                                       |
| 2 ms                           | 36%                         | 25–95%                       | 67%                                           |
| 5 ms                           | 14%                         | 10–38%                       | 22%                                           |
| 10 ms                          | 7%                          | 5–19%                        | 11%                                           |
| 25 ms                          | 3%                          | 2–8%                         | 4%                                            |
| 50 ms                          | 1%                          | 1–4%                         | 2%                                            |
| 100 ms                         | 0.7%                        | 0.5–2%                       | 1%                                            |

TABLE S14

**ADDING THE LAG BACK RECOVERS THE DELIVERY TIME WHERE THE CHECK REJECTS AS WELL AS WHERE IT PASSES.** RECOVERY ERROR AS A PERCENTAGE OF EACH CONDITION'S MEDIAN DELIVERY TIME. Q3 IS THE UPPER QUARTILE. EXACT IS THE SHARE OF CONDITIONS RECOVERED TO THE MICROSECOND THE CAMPAIGN RECORDS, WITH ITS WILSON 95% INTERVAL.

| Conditions    | $n$ | Median | Q3  | Worst | Exact (Wilson 95%) |
|---------------|-----|--------|-----|-------|--------------------|
| Check accepts | 42  | 3.4%   | 20% | 36%   | 33% [21–48]        |
| Check rejects | 28  | 12.2%  | 19% | 36%   | 21% [10–40]        |

### S3.8. The proxy's error falls as one over the delivery time, and adding the lag back repairs it

The acknowledgment lag does not grow with the path, so the transport proxy's relative error is  $A/D$  and a short path inherits it whole. Table S13 tabulates the curve of Figure 4 at nine path lengths, from the measured lags. At the median lag the error passes 100% below 0.72 ms, and there an acknowledgment-referenced harness stops measuring the path at all. The band column is the one to read, because a lag is not a constant. A 10 ms path sits at 7% at the median lag, but a tenth of our conditions sit at 19% or more. A send-referenced harness reports the delivery time itself and needs no correction from the table.

The two headline ratios of Section III-D are medians over conditions of the median of  $S$  over the median of  $D$ , and the estimator decides them. A reader who differences the per-condition medians gets an understatement of  $3.5\times$ , and one who divides the pooled medians gets  $2.6\times$ : three defensible readings, a factor of 1.6 apart. The independence overshoot of Section III-A is a median of ratios too, taken within runs above a 0.1% floor, where the ratio of medians nearly agrees with it ( $7.0\times$  against  $7.4\times$ ).

The understatement is a median, and half the conditions are worse. Its interquartile range runs  $4.0\text{--}7.5\times$ , and the worst condition reaches  $14\times$ . Two identical systems whose clients timestamp in different places differ by the difference of their lags, an apparent gap of 11% at a delivery time of 10 ms, with the sign check firing at neither. Over the 24 matched workloads of Section III-D the proxy inflates the gap between the brokers 1.7-fold, with an interquartile range of 1.2–2.0. On 21% of the pairs it narrows the gap instead.

The repair is Equation 1 read backwards. The identity holds per event and  $A$  is measured on the producer's side of one clock, so a harness that records its own lag recovers the delivery time it failed to measure. Table S14 gives the error of that recovery, as a percentage of each condition's median delivery time, split by the sign check's verdict.

Nothing in these data separates the two populations. The Hodges–Lehmann shift over all 1,176 pairwise differences, rejected minus accepted, is 1.2 points [95% percentile bootstrap:  $-1.8$  to  $8.3$ ]. The worst condition is the same on both sides, and the upper quartiles are within a point and a half. The medians differ because 33% of the accepted conditions recover the delivery time exactly, and a pile at zero drags a median down without moving the population. The pile is not itself a difference: the exact shares differ by 11.9 points, with a Newcombe 95% interval of  $-10.0$  to  $30.7$ . Without the exact recoveries the medians are 17.1% over 28 accepted conditions and 14.3% over 22 rejected ones, and the shift between them is 0.0 points [95% bootstrap:  $-8.0$  to  $7.1$ ].

The shapes differ to the eye (Figure S4). Counted in three bands of the error, exact, below 20% and 20% or more, the accepted conditions divide 14, 14 and 14. The rejected ones divide 6, 17 and 5, a smaller pile at zero with most of the rest below the edge. The edge is the accepted population's upper quartile, and the thirds hold for any edge above 14.3% up to it. The distributional test does not separate them either: a tie-corrected two-sample Kolmogorov–Smirnov statistic of 0.27 has  $p = 0.12$  over 20,000 permutations.

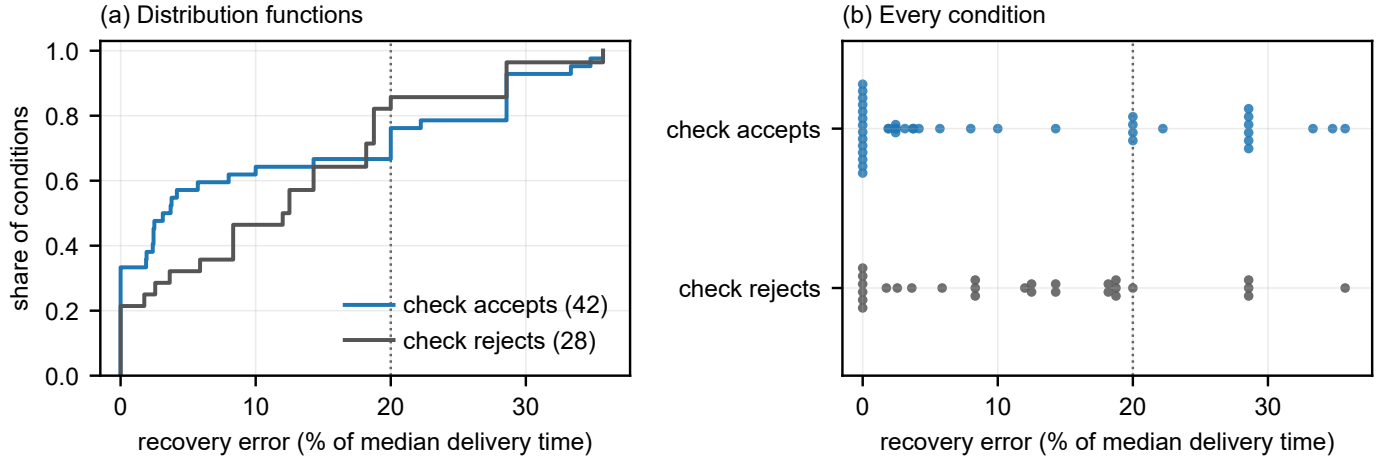


Fig. S4. **Two populations with no detectable shift between them.** (a) Empirical distribution functions of the recovery error over the 42 conditions the check accepts and the 28 it rejects. The dotted line is the band edge, the accepted population’s upper quartile of 20%. The curves open with steps of 33% and 21% at zero, change order at 18.2% and change back at 34.8%, where each population has a single condition left above. (b) Every condition, one mark each.

TABLE S15

**NONE OF FIVE OBJECTIONS MOVES A NUMBER IN THE MAIN TEXT.** ONE ROW PER OBJECTION. THE LAST COLUMN SAYS WHERE THE FULL ANSWER IS, IN THIS SUPPLEMENT OR IN THE POSTMORTEM.

| Objection                                                                                              | Answer                                                                                                                                                                                                                        | Full answer                     |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Is a negative span RFC 1242’s negative latency?                                                        | No: RFC 1242’s negative latency [S5] is a cut-through forwarding artifact, while a negative span is two host timestamps subtracted with one of them written late (Equation 3).                                                | Section II-A                    |
| Does retention mean Kafka’s log-retention setting?                                                     | No: retention here is the fraction of the samples a tool takes that survive into what it reports, a property of the measurement.                                                                                              | Section II                      |
| Why the wall clock and not <code>CLOCK_MONOTONIC</code> , and is the TSC coherent across virtual CPUs? | The wall clock gives two hosts a common epoch, the ordering a monotonic clock would guarantee on one host is shown by the data, and the probe admits only clocksources the kernel uses where cross-CPU coherence is asserted. | S3.6; postmortem, S15 and S16.6 |
| Is the two-host harness’s sample count a repeated figure?                                              | No: the two-host campaign was matched run for run to the one-clock campaign, so both hold 905,040 samples, and its 0 negatives are its own count.                                                                             | S4.6; postmortem, S22.1         |
| Did the kernel tracer change the rate it was used to explain?                                          | Yes, from 0.272 untraced to 0.231 traced ( $z = 3.6$ ), so each prediction is held against the traced run’s own rate, and the real-time configurations’ zeros are the tracer’s.                                               | S3.5                            |

The data support no equivalence either. The 90% bootstrap intervals on the two shifts would support it only against margins of 8.3 points over all conditions and 7.1 over the non-zero ones, about half the non-zero median errors. The claim is therefore the weaker one: the repair holds where the check rejects as well as where it passes. The median is also no bound. For a quarter of the accepted conditions the error reaches 20%, and for one it reaches 36%.

### S3.9. Five objections, each answered in one line

None of the five objections in Table S15 moves a number in the main text, and each has a one-sentence answer. The table gives that answer and says where the full one is. The interpreter lock stays in Section VIII, and S3.6 answers it in full.

## S4. FAILURE 2: THE FULL EVIDENCE

The benchmark’s printed latency describes only the samples its send schedule lets through, and each link in that claim can be checked against a source line, a counter or a registered prediction. Every benchmark run below is in embedded mode, with driver and workers in one JVM.

### S4.1. Two source lines make the filter, and counting what they drop changes no printed number

The positivity filter can be counted without being disturbed. In `LocalWorker.java:294` the benchmark computes its end-to-end span as

```
MILLISECONDS.toMicros(now - publishTimestamp)
```

from two millisecond timestamps, so the microseconds decorate a millisecond subtraction. In `WorkerStats.java:95` the positivity filter,

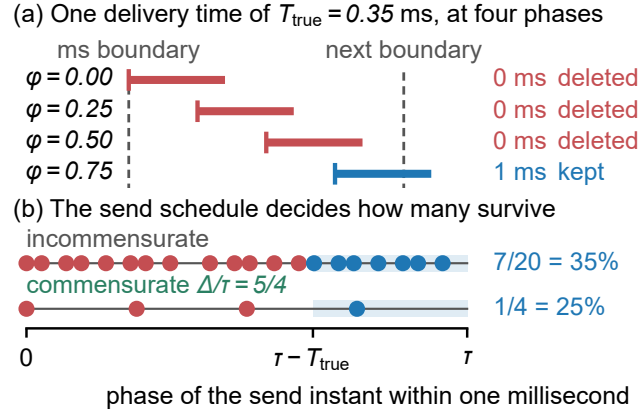


Fig. S5. **Where a send falls within a millisecond decides whether its sample survives.** (a) One delivery time of  $T_{\text{true}} = 0.35$  ms at four phases of one millisecond: three finish inside it, compute to zero and are removed; the fourth crosses the boundary and is recorded as one whole millisecond. (b) A sample survives from phase  $\tau - T_{\text{true}}$  onward, the shaded band, so uniform phases retain  $T_{\text{true}}/\tau$ ; a commensurate producer visits only  $q$  phases and retains a count out of  $q$ . The delivery time drawn is off-vertex by choice: at 0.5 ms the two answers coincide.

```
if (endToEndLatencyMicros > 0)
```

decides whether that value reaches the histogram. The message has already been counted as received, but nothing counts what the condition drops.

We add counters in an `else` branch of that condition and change nothing else: not the latency computation, not the filter, not any reported statistic. The benchmark's output is therefore exactly what it would have been unpatched, and its printed summary can be set beside the retention behind it. The four counters, for zero, negative, most negative and kept samples, separate sign. A negative span and a sub-millisecond delivery time both fail a  $> 0$  test, and one unsigned total cannot tell them apart (S4.5).

The counts are exact: the counters are shared across the JVM and their totals printed once at shutdown. A run that fails discards nothing, so its zero would look measured, and it is written as invalid, with its reason and no count. A matched sweep with the benchmark's one-minute warmup disabled leaves the retained fraction unchanged. The campaign totals 223 instrumented runs, swept one axis at a time with background load, message size, producer rate and warmup otherwise fixed.

#### S4.2. The phase of the send schedule sets retention, as a 1970 counter note already showed

Where each send falls within a step of the clock decides whether its sample survives. A delivery time shorter than the step  $\tau$  survives only if it crosses a grid instant, where the timestamp changes value. Otherwise both timestamps floor alike, and the positivity filter deletes the zero (Fig. S5a). Uniform phases retain  $\theta = T_{\text{true}}/\tau$ , which is Equation 4 of the paper.

A paced producer does not draw its phases uniformly. If its send interval  $T_{\text{send}}$  satisfies  $T_{\text{send}}/\tau = a/q$  in lowest terms, it visits only  $q$  distinct phases, so its retention is a whole count out of  $q$  (Fig. S5b). Call the retentions  $0, 1/q, \dots, 1$  the vertices of the configuration's  $q$ -grid, and the interval between neighboring vertices,  $100/q$  percentage points wide, a grid step. Replicates of one configuration land on the vertices that bracket  $\theta$ , and their spread obeys

$$\text{spread} \rightarrow \begin{cases} 100/q & \text{when } \theta \text{ sits mid-step,} \\ 0 & \text{when } \theta \text{ sits on a vertex.} \end{cases} \quad (\text{S3})$$

The step width is thus an upper bound on the spread, reached only when replicates realize both bracketing vertices. The incommensurate rates put  $T_{\text{true}}$  at 0.49 ms, half a millisecond to within replicate noise (Section IV-C). A half lies on every  $q$ -grid with even  $q$  and on none with odd  $q$ , so the rule predicts full step width for odd denominators and almost none for even ones.

None of this arithmetic is new. Hewlett-Packard's Application Note 162-1 [S6] carries it for a counter timing an interval, point by point.

- The retention identity. For an interval of  $Q + F$  clock periods the note gives  $\Pr[R = Q+1] = F$ : the share of readings on the upper of the two straddling counts is the fractional part. That is Equation 4.
- The class of a repetition rate. The class is the denominator of the ratio of repetition rate to clock rate in lowest terms, our  $q$ , and it fixes how many distinct readings the counter can produce.
- The ceiling on averaging. Averaging  $N$  readings can improve resolution by at most a factor  $1/q$ , not  $1/\sqrt{N}$ . The note names  $q = 1$  as the corner where averaging cannot help at all, and a paced producer on a round interval walks straight into it (Table S18).

TABLE S16

**REPLICATE SPREAD FOLLOWS THE PHASE DENOMINATOR  $q$  IN 10 OF THE 12 COMMENSURATE CONFIGURATIONS.** PAYLOAD, LOAD, DURATION AND HOST ARE FIXED. THE POSITION IS THE DISTANCE FROM THE MEASURED CONTINUOUS VALUE TO THE NEAREST VERTEX, AGAINST HALF THE STEP WIDTH. A CONFIGURATION IS PREDICTED FULL WHERE ITS POSITION EXCEEDS A HALF, AND SHOWS FULL WHERE ITS SPREAD EXCEEDS HALF THE STEP WIDTH. THE DAGGERED ROWS ARE THE TWO MISSES, RESOLVED IN TABLE S17. INCOMMENSURATE CONFIGURATIONS HAVE NO GRID STEP; THEIR SPREADS OF ABOUT THREE POINTS ARE THE NOISE FLOOR.

| rate                                                     | $T_{\text{send}}/\tau$ | $q$ | $n$ | width | pos. | predicts    | spread | shows       |
|----------------------------------------------------------|------------------------|-----|-----|-------|------|-------------|--------|-------------|
| 1000/s                                                   | 1/1                    | 1   | 5   | 100.0 | 0.98 | full        | 99.4   | full        |
| 500/s                                                    | 2/1                    | 1   | 5   | 100.0 | 0.98 | full        | 99.5   | full        |
| 250/s                                                    | 4/1                    | 1   | 5   | 100.0 | 0.98 | full        | 98.7   | full        |
| 400/s                                                    | 5/2                    | 2   | 5   | 50.0  | 0.04 | flat        | 17.6   | flat        |
| 600/s                                                    | 5/3                    | 3   | 6   | 33.3  | 0.95 | full        | 32.7   | full        |
| 300/s                                                    | 10/3                   | 3   | 10  | 33.3  | 0.95 | full        | 31.2   | full        |
| 800/s                                                    | 5/4                    | 4   | 5   | 25.0  | 0.07 | flat        | 7.2    | flat        |
| 1250/s <sup>†</sup>                                      | 4/5                    | 5   | 5   | 20.0  | 0.91 | full        | 7.6    | <b>flat</b> |
| 625/s                                                    | 8/5                    | 5   | 10  | 20.0  | 0.91 | full        | 19.3   | full        |
| 875/s                                                    | 8/7                    | 7   | 5   | 14.3  | 0.87 | full        | 10.7   | full        |
| 700/s <sup>†</sup>                                       | 10/7                   | 7   | 5   | 14.3  | 0.87 | full        | 6.5    | <b>flat</b> |
| 900/s                                                    | 10/9                   | 9   | 5   | 11.1  | 0.84 | full        | 6.9    | full        |
| <i>incommensurate — phase set effectively continuous</i> |                        |     |     |       |      |             |        |             |
| 333/s                                                    | 1000/333               | >64 | 5   | —     | —    | $\approx 0$ | 4.5    | —           |
| 383/s                                                    | 1000/383               | >64 | 5   | —     | —    | $\approx 0$ | 2.8    | —           |
| 457/s                                                    | 1000/457               | >64 | 5   | —     | —    | $\approx 0$ | 2.1    | —           |
| 611/s                                                    | 1000/611               | >64 | 5   | —     | —    | $\approx 0$ | 3.1    | —           |
| 889/s                                                    | 1000/889               | >64 | 5   | —     | —    | $\approx 0$ | 1.7    | —           |
| 1053/s                                                   | 1000/1053              | >64 | 4   | —     | —    | $\approx 0$ | 3.0    | —           |
| 1219/s                                                   | 1000/1219              | >64 | 4   | —     | —    | $\approx 0$ | 4.5    | —           |

- The symptom. A reading appears to hang on a round number while the input is known to vary.
- The cure. Phase jitter on the repetition rate, so the samples stop landing on the same points of the clock’s cycle. Hewlett-Packard patented it: US 3,938,042, filed in 1974 and granted in 1976, varies the phase of the counter’s own time base “to destroy coherence between the clock signal and an unknown signal” [S7]. A benchmark’s time base belongs to the operating system, so the benchmark must dither the send instant instead.

ADC metrology standardizes the phase lemma. IEEE Std 1241-2010 requires the record of its histogram test to hold a number of input cycles “relatively prime” to the number of samples (§6.4.1), and names the practice coherent sampling (§5.4.1) [S8]. Neither source contains the deletion. The counter keeps both readings, and the retained fraction is its estimator.

#### S4.3. Retention sits on the vertices in nine of the ten configurations able to show it

The phase account predicts each configuration’s behavior from its denominator alone, and every configuration a test can judge falls on the side of its  $q$ -grid. Publish latency cannot be what moves retention. It takes two values, 0.3 and 0.4 ms, across settings whose retention varies 279-fold. The correlation it does have (Spearman +0.31; Fisher 95%: +0.08 to +0.51; 71 settings) carries the sign the retention law predicts.

Table S16 tests the spread rule configuration by configuration. A spread above half the step width counts as full, so the test needs no tolerance. Of the 12 commensurate configurations, 10 match, and the predictions split ten full against two flat. The two misses, 700 and 1250 msg/s, are predicted full and measured flat. Spread is a range over replicates of a quantity with two-point support, so a configuration shows its full width only if its replicates land on both bracketing vertices. At  $q = 5$  and  $q = 7$  the upper vertex carries a minority of the probability, and five replicates can all fall on the lower one however exact the grid is. Spread cannot tell that outcome from a refutation.

We built a replacement statistic for that reason after the campaign had run; it was not registered, and its first results were committed with it. For each configuration,  $\bar{d}$  is the mean distance of its replicates to the nearest vertex of its  $q$ -grid, divided by the step width. The null is a continuum: replicates drawn from a Normal distribution centered on the configuration’s rate-local continuous value, with a measured standard deviation. Neither parameter is fitted to the configuration under test. The one-sided  $p$ -value comes from 4,000 Monte Carlo draws, which puts its floor at  $2.5 \times 10^{-4}$  before correction. It simulates the null at the observed number of replicates rather than leaning on an asymptotic distribution, which matters at five.

Of the 12 configurations, ten are powered, meaning the null and the grid predict separably. Of these, nine reject the continuum null at  $\alpha = 0.05$  after Holm correction, eight of them at the Monte Carlo floor (Table S17, Fig. S6). The tenth, 900 msg/s, rejects before correction ( $p = 0.044$ ) and not after ( $p = 0.131$ ), so we report it as unresolved. The two unpowered configurations, 400 and 800 msg/s, have grids that pass through  $\theta$ , where the two hypotheses coincide; the test reports them as undecidable. Both configurations the spread rule missed reject the continuum: 700 msg/s at 0.009 and 1250 msg/s at 0.003.

Two rates that share a denominator land on the same vertices. The registered control is 600 msg/s, an interval of 5/3 ms and so the same  $q = 3$  as 300 msg/s, registered in advance to land near 33%. Its replicates were 33.98, 34.29, 34.90, 36.17,

TABLE S17

**RETENTION SITS ON THE  $q$ -GRID IN NINE OF THE TEN POWERED CONFIGURATIONS, TESTED AGAINST A CONTINUUM NULL.** HERE  $\bar{d}$  IS THE MEAN NORMALIZED DISTANCE TO THE NEAREST VERTEX, AND THE NULL COLUMN IS ITS MEAN UNDER THE SIMULATED NULL. THE COLUMN  $p_{\text{Holm}}$  IS A MONTE CARLO  $p$ -VALUE (4,000 DRAWS), HOLM-CORRECTED WITHIN ALL 12 CONFIGURATIONS; IT DECIDES THE VERDICT. FLAT (COINCIDENT) MARKS CONFIGURATIONS WHERE THE TWO HYPOTHESES PREDICT ALIKE AND THE TEST HAS NO POWER.

| rate   | $a/q$ | $n$ | $\bar{d}$ | null  | $p_{\text{Holm}}$ | verdict           |
|--------|-------|-----|-----------|-------|-------------------|-------------------|
| 1250/s | 4/5   | 5   | 0.131     | 0.351 | 0.003             | grid              |
| 1000/s | 1/1   | 5   | 0.003     | 0.479 | 0.003             | grid              |
| 900/s  | 10/9  | 5   | 0.265     | 0.340 | 0.131             | not resolved      |
| 875/s  | 8/7   | 5   | 0.209     | 0.382 | 0.003             | grid              |
| 800/s  | 5/4   | 5   | 0.104     | 0.058 | 1.000             | flat (coincident) |
| 700/s  | 10/7  | 5   | 0.329     | 0.416 | 0.009             | grid              |
| 625/s  | 8/5   | 10  | 0.112     | 0.447 | 0.003             | grid              |
| 600/s  | 5/3   | 6   | 0.034     | 0.469 | 0.003             | grid              |
| 500/s  | 2/1   | 5   | 0.043     | 0.490 | 0.003             | grid              |
| 400/s  | 5/2   | 5   | 0.093     | 0.020 | 1.000             | flat (coincident) |
| 300/s  | 10/3  | 10  | 0.221     | 0.468 | 0.003             | grid              |
| 250/s  | 4/1   | 5   | 0.181     | 0.489 | 0.003             | grid              |

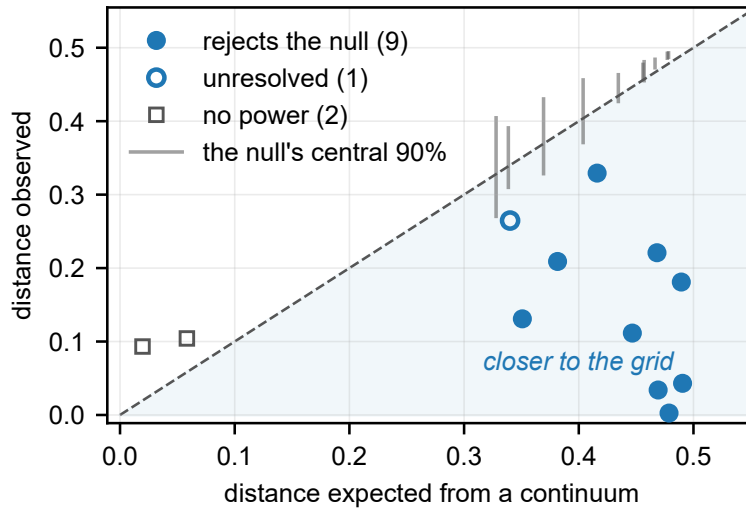


Fig. S6. **All ten powered configurations fall on the grid side of a continuum, and nine survive correction.** Observed mean distance to the nearest vertex against the distance a continuum would give, the diagonal. Each gray bar is that configuration's simulated null, uncorrected, and a marker below its bar rejects it. Filled circles also survive Holm correction. The open circle, 900 msg/s, sits below its bar, 0.2647 against 0.2680, but not after correction ( $p = 0.131$ ). The two squares have no power and no bar.

65.86 and 66.67: four on the  $1/3$  vertex and two on the  $2/3$  vertex, with nothing between. Their median, 35.5%, is near the 33% registered, and none left the thirds grid, the registered falsifier. The upper two sit on the  $2/3$  vertex that 300 msg/s also reaches, an octave away. Table S18 shows the  $q = 1$  corner replicate by replicate.

The registered test that manipulates the physics is the payload flip. Payload moves  $T_{\text{true}}$ , and so moves  $\theta$  within its grid step, with nothing else changed. Measured at an incommensurate rate (Table S19), 32 KB puts  $\theta$  almost on the  $2/3$  vertex of the  $q = 3$  grid and 64 KB puts it mid-step. The prediction was that this one manipulation would move a 300 msg/s configuration onto a vertex and off it again. The flat-full flip came true in both directions (Fig. S7). At 32 KB the five replicates spanned 62.06–75.69%, a spread of 13.6 points, with three agreeing to 0.06. At 64 KB they spanned 73.06–99.78%, a spread of 26.7 points. Both cross the flat-full boundary at half the step width, 16.7 points, in the predicted direction. One registered detail

TABLE S18

**ONLY THE COMMENSURATE RATE IS UNSTABLE.** RETENTION AGAINST THE PRODUCER'S SEND INTERVAL, WITH MESSAGE SIZE, BACKGROUND LOAD, DURATION AND HOST FIXED. AT 500 MSG/S THE INTERVAL IS EXACTLY TWO MILLISECONDS, SO  $q = 1$ . THE TWO INCOMMENSURATE RATES AGREE WITH EACH OTHER ALTHOUGH THEIR INTERVALS DIFFER BY 19%. PREDICTIONS WERE RECORDED BEFORE THE RUNS.

| Rate  | Interval | Multiple of 1 ms? | Retention per replicate           | Spread |
|-------|----------|-------------------|-----------------------------------|--------|
| 500/s | 2.000 ms | exact             | 0.47, 1.51, 1.55, 18.02, 99.99    | 99.5   |
| 457/s | 2.188 ms | no                | 48.77, 49.50, 49.69, 50.51, 50.87 | 2.1    |
| 383/s | 2.611 ms | no                | 50.28, 51.09, 51.45, 51.77, 53.13 | 2.8    |

TABLE S19

AT A RATE INCOMMENSURATE WITH THE MILLISECOND, RETENTION READS OFF THE DELIVERY TIME. RETENTION AGAINST PAYLOAD AT 457 MSG/S, WHERE SAMPLES STAY DEPHASED. THE IMPLIED  $T_{\text{true}}$  INVERTS EQUATION 4. UP TO 8 KB THE ADDED DELAY IS COMPARABLE TO THE REPLICATE NOISE; THE 32 AND 64 KB LEVELS CARRY THE RESULT.

| Payload | Retention (3 reps)  | Median | Implied $T_{\text{true}}$ |
|---------|---------------------|--------|---------------------------|
| 200 B   | 51.08, 52.03, 52.88 | 52.03% | 0.520 ms                  |
| 1 KB    | 50.11, 50.32, 51.47 | 50.32% | 0.503 ms                  |
| 2 KB    | 52.13, 54.62, 54.77 | 54.62% | 0.546 ms                  |
| 4 KB    | 51.86, 54.17, 56.24 | 54.17% | 0.542 ms                  |
| 8 KB    | 54.92, 55.22, 57.89 | 55.22% | 0.552 ms                  |
| 32 KB   | 68.01, 68.44, 69.40 | 68.44% | 0.684 ms                  |
| 64 KB   | 84.21, 85.36, 86.82 | 85.36% | 0.854 ms                  |

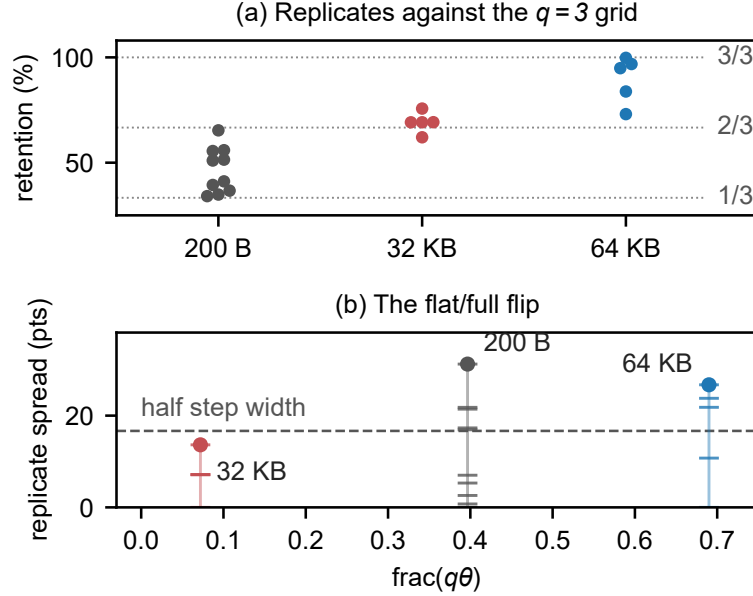


Fig. S7. **One manipulated variable moves a configuration onto its grid and off it.** (a) Replicates at 300 msg/s ( $q = 3$ ) by payload, against the thirds grid: 200 B straddles both branches, 32 KB pins near the 2/3 vertex, and 64 KB re-opens to the upper step. (b) Replicate spread against  $\text{frac}(q\theta)$ , each stick showing the replicates its spread is made of, with  $\theta$  each configuration's own mean retention in this campaign; the registered prediction placed 32 and 64 KB at 0.055 and 0.56 from the payload sweep's  $\theta$ . The flat–full boundary at half the step width is crossed in both directions.

failed: the 32 KB configuration pins at 69.2%, not on the vertex at 66.67%. The flip was one of six predictions registered with falsifiers before its campaign. It was confirmed, three falsifiers fired, one prediction could not be evaluated, and one split (postmortem, S1.7).

#### S4.4. A deleted zero changes what the distribution is conditioned on

Deleting the samples that compute to zero conditions the reported distribution on being positive, so the retained fraction is part of the measurement. Two classical sources come close to this argument. The GUM models a digital reading of resolution  $\delta x$  as a stimulus that “can lie with equal probability anywhere in the interval  $X - \delta x/2$  to  $X + \delta x/2$ ” (Clause F.2.2.1) [S9]. That rectangular model, with its variance  $(\delta x)^2/12$ , holds for retained readings. The positivity filter removes one side of the interval, and neither the shape nor the variance survives the asymmetry it adds.

Pratt showed that discarding zero differences before ranking breaks a natural monotonicity: a sample not judged significantly positive can become so when every observation is decreased [S10]. His procedure ranks all observations, zeros included, drops only the zeros' ranks, and refers the statistic to its null distribution conditional on the number of zeros. Our claim is weaker and more practical: the zero count must be published, so a reader can see what the distribution is conditioned on. The exact parallel is a reading Pratt reports from Meier, that a zero arises because “with only our crude measurement, we do not know” the sign a finer measurement would resolve. That is the sub-resolution sample of Failure 2, stated in 1959.

Figure S8 applies the rule to the population of Table I. With millisecond timestamps a positivity filter removes 338,242 of the 738,730 events, 45.8% (95% bootstrap interval, resampling runs: 45.4–46.2). Of those, 303,468 computed to exactly zero and 34,774 to less, and a harness that filters this way counts none of them. Of the five dispositions in panel (c), two shrink the sample and publish no count of what left, two hold the count fixed by reporting a value never measured, and one keeps the

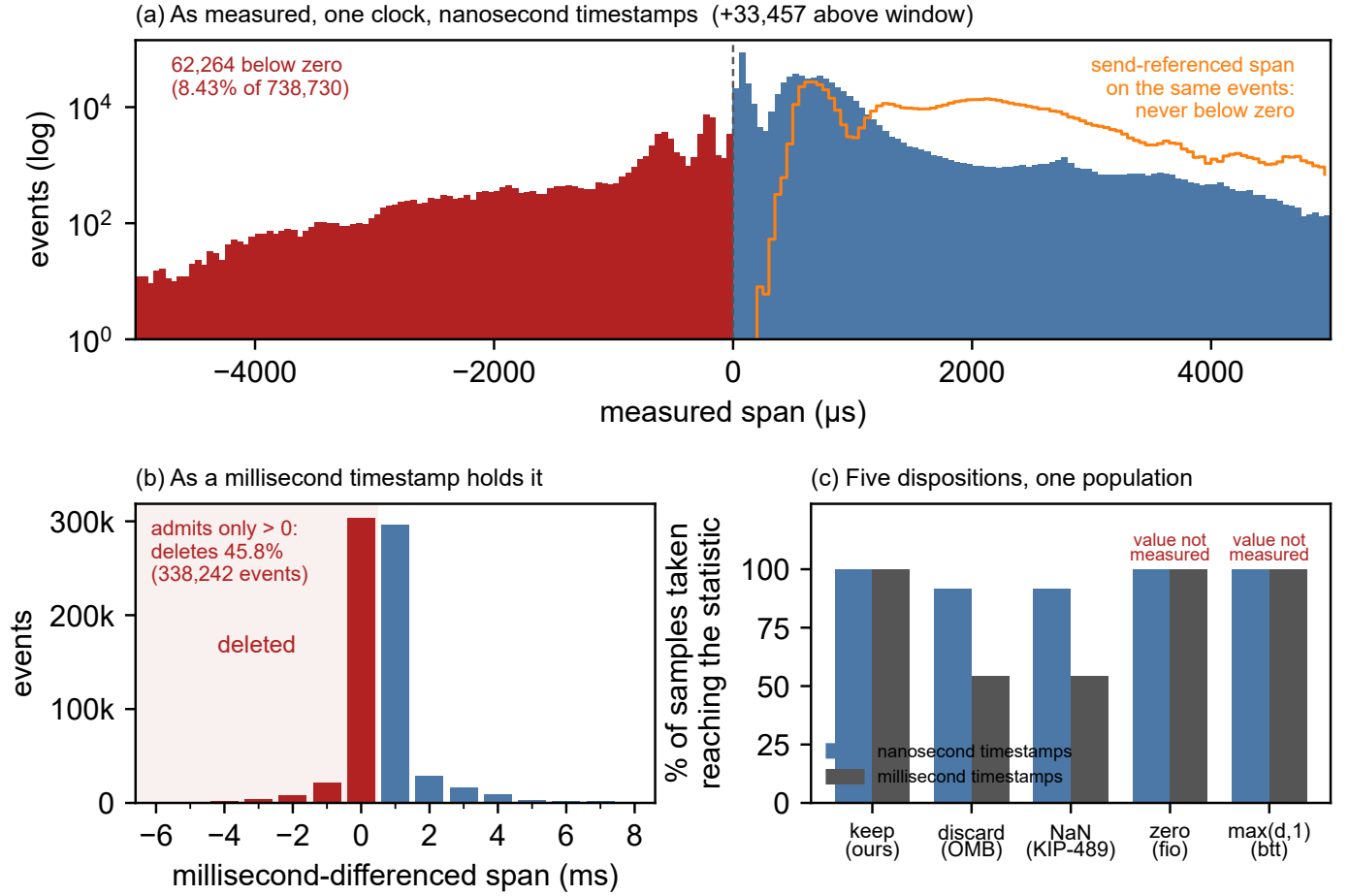


Fig. S8. **Positivity filter on millisecond timestamps would delete 45.8% of our corpus, uncounted.** (a) At nanosecond resolution the acknowledgment-referenced span goes below zero 62,264 times in 738,730; the send-referenced span, on the same events and clock, never does. (b) Differenced millisecond timestamps land on 0 or 1 by where they fall against the millisecond boundaries alone, and the filter deletes every value at or below zero. (c) Five dispositions applied to the same population.

data. This is the rule applied to our spans; what the benchmark’s own instrumented runs retained is the other measurement, in Section IV-B of the paper.

#### S4.5. Two floored clocks disagree by exactly one millisecond, so the sign of a discard names the failed clock

The sign of a discarded sample says which clock failed, and the benchmark’s two drivers show both kinds. The Kafka driver’s 214 runs discarded 10,913,263 samples, and not one was negative. The Redis-driver replication’s 9 runs carry all 41,403 of the campaign’s negatives, every one exactly  $-1000 \mu$ s. A single run holds 41,397 of them, half of all the samples it took, beside six kept.

The drivers explain the difference. The Redis driver takes its publish timestamp from `entry.getID().getTime()` in `RedisBenchmarkConsumer.java:72`, the stream-entry identifier the Redis server assigns on `XADD`. The Kafka driver uses a producer-set timestamp. So the Redis span subtracts a server-side timestamp from a consumer-side clock reading, across two hosts, with both floored to the millisecond. The Kafka span floors two timestamps whose difference is taken inside one JVM.

Two consequences follow. First, two clocks floored to the same resolution and offset by less than one millisecond can disagree by one millisecond and by no other amount. The only negative a floored cross-host difference can produce is therefore  $-1000 \mu$ s, and every one of the corpus’s 41,403 negatives is that value. Clock discipline was clean at every logged minute, so this failure needs no misbehaving clock, only two clocks and a floor. Second, a difference taken inside one process has no second clock to disagree with, which is why the Kafka discards hold no negative. The same positivity filter thus hides a resolution failure under Kafka and a clock failure under Redis, and an unsigned discard counter could not tell them apart. The two drivers also do not share a measurand: client to client on one clock under Kafka, server to client across two under Redis. The benchmark does not flag the difference.

#### S4.6. An independent harness reproduces the deletion, and across two hosts the loss follows the clocks

The deletion does not depend on the benchmark’s code, and across two hosts the deleted fraction drifts with the clocks. We reproduced the positivity filter in a Python harness that shares no code with the benchmark, with the same timestamp, subtraction and filter. On one clock it recorded 0 negative differences across 905,040 samples in 12 runs, as a causal chain requires, and the grid reappeared intact. Across two hosts with disciplined clocks, 12 runs again recorded 0 negatives in 905,040 samples and an intact grid, but the deleted fraction followed the drift of the clock offset. At 457 msg/s, the incommensurate configuration and the only one of the three whose own grid spread does not swamp the comparison, cross-host retention wandered from 13.4 to 26.9% over four replicates. On one clock the same configuration held to 0.98 points. Nothing records either the synchronization state or the fraction it deletes.

The benchmark’s own distributed mode, where its subtraction spans two clocks, could not be measured. It never completed a run in eleven attempts: eight at three background-load levels, including none, and three more run with full diagnostics. The unloaded attempts failed like the loaded ones, which refutes our own hypothesis that processor starvation was the cause. The three diagnosed attempts failed identically, in the coordinator-to-worker path before any workload started. The first attempt, made before the validity gate of S4.1 existed, wrote a vacuous zero that the artifact keeps marked invalid; the gate refused every later attempt, each with its reason. The exposure of the distributed binary therefore rests on reading its source: the timestamp is written on the producer’s host and read on the consumer’s, and the positivity filter drops whatever the difference produces.

#### S4.7. A patched benchmark still reads the brokers’ millisecond timestamps

Repairing the benchmark does not reach the brokers’ own timestamps, because both brokers write them in milliseconds. Kafka’s `record.timestamp()` is a millisecond field on the wire. Anything built on it inherits a millisecond resolution and must carry its own finer timestamp in the payload to escape. Documenting that, or exposing a finer timestamp attached to the record, is the change to ask of Kafka. The Redis XADD entry identifier embeds a server-clock time floored to the millisecond. One documented sentence saying it is not comparable with a client clock would have prevented the driver defect of S4.5. No layer here misbehaves. The benchmark is the only party that assumes something none of them promised: that observing an event tells you when the event happened. For benchmark authors the checks are those of Table IV, and one more: never difference a server-assigned timestamp against a client clock without saying so.

### S5. THE TOOL REGISTRY

Five of the ten tools we read at source dispose of a non-positive or sub-resolution sample without counting it, and every verdict traces to a line of source a reader can check. Each row of the registry file carries one source line, the file and symbol it came from, the upstream URL and the date it was read. The class labels are not stored. They are recomputed on every build and every test run by the classifier that read the OpenMessaging Benchmark a month before the other tools were audited. A reader who disagrees with a verdict can therefore check the line against the pattern that matched it. The classifier sees only what its patterns recognize. They were written against JVM sources and had to be extended before they could read an Erlang guard or clock call, so the registry bounds the field from below. We call a tool’s own conditional on the sign of an interval its *guard*. A guard is a deletion only if what it guards is the sample. Table III of the paper is the registry, one row per span because one tool can treat two spans differently, generated from the registry file at build time; its sources, in row order, are [S11], [S11]–[S14], [S14]–[S20].

The five silent disposals take three forms. A filter drops what its comparison excludes:  $> 0$  deletes every sample that computes to exactly zero, while  $\geq 0$  keeps those and drops only negative spans. A filter shrinks the sample the statistic is computed from and publishes no count of what left. The benchmark filters at  $> 0$ . Apache Pulsar’s performance client guards at  $\geq 0$ , so it drops negative spans and keeps the zeros the benchmark deletes, which then pile up visibly at zero. A substitution keeps the sample count, so any retention rate computed downstream reads 100%, which makes it harder to detect than a filter. A library refusal leaves the sample unrecorded because nobody checks the library’s answer. Truncation is a separate loss that keeps every sample. The benchmark truncates before it filters, and Kafka’s bundled tools only truncate. The notes below carry what the table cannot.

- The OpenMessaging Benchmark appears twice because it treats its two spans differently (Section IV-A).
- The `EndToEndLatency` tool bundled with Kafka cannot invert: it uses one thread, one monotonic clock and a synchronous send. It loses the measurement anyway. Each sample prints in floating point, but the `long[]` its percentiles come from is filled by the integer division `elapsed / 1000 / 1000`. On a sub-millisecond path its p50, p99 and p999 therefore all read zero. The companion `ProducerPerformance` takes its send-to-acknowledgment difference on `System.currentTimeMillis()` and casts it to `int`.
- In `fio` the guard appears in three timing functions, each returning zero for a negative interval under the comment “time warp bug on some kernels?”. The behavior is defensive and reasonable in isolation; what is missing is any record that it fired.
- In `btt`, `tdelta()` returns one nanosecond whenever the stage timestamps are out of order, with no comment, no counter and no mention in the manual page.

- The `wrk2` load generator detects the negative interval, prints a block that begins “We are about to crash and die”, and records the value anyway without checking the result. It builds against its own copy of `hdr_histogram.c`, which bounds-checks the derived index and returns an unchecked `false`, and whose `assert()`s are live in the shipped build. Either the assertion aborts the run or the refusal drops the sample, and nothing counts the loss.
- In `emqtt-bench` the `> 0` guard protects a counter and leaves the histogram alone, so nothing is deleted from the sample.
- RabbitMQ PerfTest and the NATS CLI pass a negative span into their metrics with no sign test at all, as RFC 7679 asks of a measurement. Both are exposed to clock skew between hosts.
- Rezolus counts its discards in a metric that names the cause.
- Two further tools do not make the measurement. SPECjms2007 defines Delivery Time and never reports it, and Redpanda’s `rpck` ships no latency benchmark.

The same disposal has reached the broker itself. In November 2025 Kafka’s coordinator metrics were found throwing on negative durations after a host clock stepped backwards, and the merged fix clamps the negative to zero with no counter and no log line [S21]. That is silent substitution, adopted in the system whose benchmark opened this audit. Its cause is genuine wall-clock non-monotonicity, a different mechanism from Failure 1. The same project makes the opposite choice a few directories away. Kafka Streams’ end-to-end latency metric records the wall clock less the record timestamp with no filter, so a record stamped in the future is recorded as a negative latency [S22]. One project clamps the negative in one metric and keeps it in another, so the disposal is a choice.

The positivity filter persists downstream and upstream. Of the 40 readable public forks of the benchmark surveyed on 2026-08-24, 40 carry the filter’s expression unchanged. That bounds how few have diverged and says nothing about why. Confluent’s fork, one of the vendor distributions practitioners run, was archived by its owner on 28 February 2026 and is read-only, with 60 forks of its own (read 2026-09-12). A live fork may drop the filter tomorrow; an archived one cannot, and neither can anything downstream of it. Upstream, a search of the benchmark’s issue tracker for the variable the filter tests, run on 2026-09-13, returns 1 item: pull request #56, which introduced the filter [S23]. No issue or pull request has named it in the eight years since. That is why the checks of Table IV are addressed to benchmark authors and to their readers.

Other runtimes document the structure behind Failure 1, a timestamp written when a thread next runs. The `librdkafka` client library serves its delivery-report callback from `rd_kafka_poll()` and the other queue-serving calls [S24]. The callback therefore runs on the application’s own thread, at whatever moment that thread next polls, rather than when the broker’s acknowledgment arrives. The gap between the two plays the part of  $\delta_{\text{ack}}$  in Figure 2. The `fio` documentation defines completion latency for asynchronous ioengines as running to “the time ... when the I/O’s completion was reaped by `fio`” [S25], so a reaping timestamp is written into the definition of the metric. Neither frames this as a hazard. The property is public, and its consequence for a subtraction across two processes is drawn nowhere we could find. An origin timestamp written by a thread that must first be woken is late by that wait, and the span it opens inverts whenever the wait outlasts the delivery time.

### S5.1. Few published results carry the deletion’s signature, and none states what it kept

Published results rarely carry the deletion’s signature, and no report we found says how many samples it kept. The audit behind that sentence was registered, committed and public before any report was read. It fixed the signature, the places to search, the rule for inclusion, the coding and the outcomes, and nothing in it changed afterwards. The primary signature is a printed end-to-end median of exactly 1 ms, the smallest value the benchmark can print and the value every retained sample takes when the true delivery time is shorter than the resolution. The strong signature adds a p99 of exactly 1 ms. Neither says a published number is wrong. Each says the number came from a fraction of the samples that the report does not state.

The search covered Google Scholar, Semantic Scholar, the web, the engineering blogs of the vendors whose brokers the benchmark has drivers for, and the benchmark’s repository and its public copies, for reports dated up to 28 September 2026. A report was included if it named the benchmark or a fork of it as its tool and printed at least one numeric end-to-end percentile. The audit included 43 reports, printing 1,100 configurations, of which 868 print a median. The primary signature occurs in eight configurations in two reports, both public repository text. Five are Redis rates in a notebook behind a 2023 journal paper that shows its medians only in a chart, and three are maximum-rate runs in three result files of a 2021 fork. The strong signature occurs nowhere, and no included report states a retained fraction or a discard count. The rules did not foresee several patterns. 14 of the included reports print medians below a millisecond, which the upstream clock cannot produce, among them vendors’ copies whose maintainers changed how the span is timed. One 2024 fork prints p95, p99 and p999 all at exactly 1 ms beside an average just above it and no median column, so by the registered rule it does not count. Four configurations print a median of zero, from empty histograms. The coding, the search log and every excluded report with its reason are in the artifact (postmortem, S25.1).

## S6. THE REGISTERED LAW CAMPAIGN

A law for the negative-span rate of Failure 1, fixed before its runs, had one prediction confirmed on every campaign, three on one broker, and the rest not. What held on every campaign that tested it, five on Kafka across the three pairs, is that

load raises the plateau. No test of whether the fall follows a slice set by hand was confirmed. Frozen programs judged every prediction over 8,102 runs on machines the earlier campaigns never ran on. The paper gives the outcome in Section VIII, and the postmortem, Part V, keeps the run-by-run record.

### S6.1. The law, and the design that tested it

The registered law says that the scheduler’s settings alone fix where the negative-span rate falls as the delivery time lengthens, and how wide the fall is. It takes the wait whole: a timestamping thread kept waiting waits out a slice  $s$  of another task and then up to one tick  $h$ . The rate then sits on a plateau  $c + p$  while the delivery time  $D$  is shorter than the slice, falls across one tick, and rests on a floor  $c$  beyond it:

$$r(D) = c + p \cdot \begin{cases} 1, & D \leq s, \\ (s + h - D)/h, & s < D < s + h, \\ 0, & D \geq s + h. \end{cases} \quad (S4)$$

The slice and the tick predict where the fall starts and how wide it is; the plateau’s height and the floor are measured. In the plan, a reading is negative when the wait of the thread that timestamps the acknowledgment outlasts the margin  $D - g$ , a run’s median delivery time less its median acknowledgment lag  $g$ .

Three pairs of Microsoft Azure machines in one region ran the campaign, each a driver with eight virtual CPUs and a broker with two; two pairs have AMD EPYC 9V74 processors and one has Arm Neoverse N2. The plan’s blocks A1 to A9 are written L1 to L9 here. L1 sets the slice on the first x86 pair, and L4 on the Arm pair. L2 sets the tick, on three kernels built from one source tree that differ in the tick alone: 1, 4 and 10 ms. L3 sets the load, L5 the number of online CPUs, L7 real-time priority for the timestamping threads, and L8 the client, Kafka’s own Java client against our Python one. L6 records every `python3` thread’s waits in half of L1’s and L3’s runs, and L9 records each of those threads’ scheduling events in half its own runs. M0 asks why, when messages come in bursts, the delivery time does not grow one-for-one with the added delay.

Each run replays messages at a steady 50 a second, or a football match’s bursts in M0, and measures its delivery time, acknowledgment lag and negative-span rate after a 30 s warm-up. The delivery time is set by delaying only the traffic to the receiver’s own network address, so the acknowledgment’s path is untouched, and each session first calibrates how far each added delay moves it. A delivery time shorter than the client’s own with nothing added cannot be set, and was not run.

The integrity rule judges every run as it ends. A run counts when every condition it was meant to have took, from the load within 3 points of its target to the added delay within 50  $\mu$ s of its setting; otherwise it is repeated, up to three attempts. A run whose median acknowledgment lag moved too far stopped its campaign. In its final form, from version 20, this lag brake held a run against its setup’s earlier runs in the campaign. The plan’s own name for the lag, and for the brake on it, is “got-it”. Each campaign that tests a prediction took its rounds, at least four and at most forty, from simulating its own decision rule on data that follow the law. L2 and L9, which have no rule the simulation can take, ran at the floor of four, M0 ran a fixed ten, and no campaign ran more than 24. A prediction about where the fall sits is read on a free curve required only never to rise and on the law’s fitted shape, and it passes only where both agree. The judging read 8,102 runs in 156 campaigns, taken from 16 to 27 September 2026, and 7,996 of them counted.

### S6.2. Every change to the plan, and what prompted it

The plan changed after results, and every change says whether one prompted it. It was first frozen as version 5 on 16 September 2026, before its first run, and the version in force at the end is the thirty-second, freeze 28. A change is never an edit. It is a new, numbered freeze, public before the runs it covers, whose first section lists what changed, why, and whether a result already seen prompted it. Table S20 gives every freeze with that answer. Versions 6 to 8 corrected what stage 0’s first runs found in the machines, the code and the plan’s own checks, before any campaign that tests a prediction began. P3 became P3a and P3b in version 9, before any load run was read. Three kinds of change followed the campaigns’ own results. Version 14 judged P3a only where the plateau reaches 2%. The lag brake was corrected five times, each time after it had stopped a campaign. And three versions corrected programs that had already judged, each stating what the verdicts it touches had been. Blocks added as the campaign ran, in versions 24 and 27 to 29, were each read by a program fixed before their runs. The last three of those versions followed a check of all the data on 25 September. P6 was given its program in version 30, at a margin no run had been read against.

### S6.3. Load raised the plateau everywhere, and the fall did not sit where the law put it

Load raised the plateau in all five load campaigns, all on Kafka, every interval above zero (Table S21 gives each verdict as its frozen program returned it). On Kafka, real-time priority took the plateau to zero in every run, and the fall followed the core count at the one count whose slice Kafka’s shortest delivery time reaches. Kafka’s own Java client shows the same fall and the same effect of priority, at a plateau a third to a quarter of the Python client’s. On the first x86 pair the fall’s halfway point lay inside its band at every slice each broker reaches on the free reading, four for Kafka and five for Redis, but on the

TABLE S20

**EACH CHANGE TO THE PLAN WAS FROZEN BEFORE THE RUNS IT COVERS, AND SAYS WHETHER A RESULT PROMPTED IT.** EVERY FREEZE OF THE LAW CAMPAIGN’S PLAN. PROMPTED: YES WHERE ANY OF THE FREEZE’S CHANGES FOLLOWED A RESULT ALREADY SEEN, WHICH IS NAMED. T1 TO T5 ARE THE PLAN’S TOOLS BLOCK. FREEZE 29, AN AUDIT OF PUBLISHED RESULTS, IS A SEPARATE STUDY.

| Freeze | Version | Frozen | What changed                                                                                                                                                                             | Prompted by a result?                                                                                    |
|--------|---------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 01     | 5       | 16 Sep | The plan as registered: the law’s predictions and the rules that judge them, the blocks, the rounds rule, the integrity rule and the lag brake                                           | No: frozen before the first run                                                                          |
| 02     | 6       | 17 Sep | The calibration must be known within 0.3 ms; the broker’s own capture checks the added delay; Redis’s consumer acknowledges in batches of 200; M0 replays bursts; stage 0 restarts       | Yes: stage 0’s first runs, which found faults in the machines, the code and two of the plan’s own checks |
| 03     | 7       | 18 Sep | Every judgment of the added delay is read from the broker’s capture; ping is recorded and decides nothing                                                                                | Yes: a sound pair turned away on a ping reading                                                          |
| 04     | 8       | 18 Sep | A slice below a client’s zero-delay delivery time is out of reach and not run, and P1 and P9 are judged on the slices a pair reaches; the lag brake must clear its session’s own scatter | Yes: the first pair’s calibration, and a spread pilot stopped by a brake set at 0.056 ms                 |
| 05     | 9       | 19 Sep | P3 becomes P3a and P3b, and P3a is judged on its whole interval; L3 runs 16 rounds                                                                                                       | No: a simulation, before any load run                                                                    |
| 06     | 10      | 20 Sep | Rounds are simulated at each broker’s own spread, never at the two pooled                                                                                                                | No: spread pilots and a simulation                                                                       |
| 07     | 11      | 20 Sep | Each L5 session measures its own baseline at its own core count                                                                                                                          | No                                                                                                       |
| 08     | 12      | 20 Sep | Tools block: seven tools’ predictions fixed from their source, and one earlier classification recorded as refuted                                                                        | Yes: the source audit refuted that classification; no run was read                                       |
| 09     | 13      | 20 Sep | P8’s program, which read a delivery time L8 never runs, is corrected; L8 runs 6 rounds; the clients refuse a clock coarser than 25 $\mu$ s                                               | No: a simulation, before any L8 run                                                                      |
| 10     | 14      | 20 Sep | P3a is judged only where the plateau reaches 2%, a lower load being out of reach; rounds are simulated at the pilots’ measured levels                                                    | Yes: L3’s first campaign, which could not place the fall at 50% load                                     |
| 11     | 15      | 21 Sep | Tools block: T2’s clock offsets follow each session’s own delivery time, and its verdict names the behavior that survives                                                                | No: a synthetic run of the block                                                                         |
| 12     | 16      | 21 Sep | The lag brake compares a run only with a baseline whose acknowledgment was timestamped the same way                                                                                      | Yes: the brake stopped L8 on the Arm pair at its first Python run                                        |
| 13     | 17      | 21 Sep | The lag brake holds a run against its campaign’s earlier runs of the same setup, not against the calibration                                                                             | Yes: the brake stopped L8 on the Arm pair at 28 of 144 runs                                              |
| 14     | 18      | 21 Sep | The lag brake’s scatter is pooled over the campaign’s setups, and waits for five degrees of freedom                                                                                      | Yes: the brake stopped L2’s first session at 28 of 40 runs                                               |
| 15     | 19      | 22 Sep | P8’s program tested two of its three clauses; the priority cut is added                                                                                                                  | Yes: found while judging L8’s Arm campaign, whose verdict did not move                                   |
| 16     | 20      | 22 Sep | The calibration’s precision bound follows the tick; the lag brake’s floor rises from 0.10 to 0.25 ms                                                                                     | Yes: three calibrations refused, and the brake stopped L8 on x86 at 86 of 144 runs                       |
| 17     | 21      | 22 Sep | Predictions are judged only on the runs the integrity rule passed; L2’s Kafka half moves to the first x86 pair                                                                           | Yes: counting L8’s x86 runs found thirteen at a point designed for twelve                                |
| 18     | 22      | 22 Sep | The calibration’s check on the acknowledgment lag is read as a slope across all its steps                                                                                                | Yes: a sound calibration ended on a +0.342 ms shift at 1 ms added                                        |
| 19     | 23      | 22 Sep | Tools block: each T2 session adds a run with no clock offset, and one tool’s two-clock setup is checked first                                                                            | Yes: the first tool runs                                                                                 |
| 20     | 24      | 24 Sep | L3’s Kafka campaign is added on the Arm pair, its Redis half left out, and P3b recorded as underpowered in advance                                                                       | Yes, for leaving Redis out: the Arm pair’s spread pilot                                                  |
| 21     | 25      | 24 Sep | Tools block: T2 is judged against its own control, and three tools that measured nothing run again; L3 on Arm is read at its measured loads                                              | Yes: the check of 24 Sep                                                                                 |
| 22     | 26      | 25 Sep | L5 at 2 CPUs finishes its last 13 runs with the lag brake only recording; T2’s count check is corrected                                                                                  | Yes: the brake stopped L5 at 2 CPUs twice                                                                |
| 23     | 27      | 25 Sep | A 3.75 ms slice for L1, second stock-kernel days for L2 and a third x86 L3 campaign; five faults of the judging programs recorded                                                        | Yes: the check of all the data, 25 Sep, including P3a read as frozen                                     |
| 24     | 28      | 25 Sep | L9 is added, with the program that reads it                                                                                                                                              | Yes: the same check, where the fitted shape failed                                                       |
| 25     | 29      | 25 Sep | M0’s recording is built, and M0 runs                                                                                                                                                     | Yes: the same check found M0 had never run                                                               |
| 26     | 30      | 25 Sep | The five faults of the judging programs are corrected, P6’s missing program among them; L9-2b is added before any L9 run                                                                 | Yes: the same check (L9-2b: no)                                                                          |
| 27     | 31      | 25 Sep | How M0 is read, fixed before any M0 run is read                                                                                                                                          | No                                                                                                       |
| 28     | 32      | 26 Sep | The lag brake only records, in every campaign; stopped campaigns are finished on their own queue, calibration and kernel                                                                 | Yes: five brake stops on 25 and 26 Sep                                                                   |

fitted shape at two of four slices for Kafka and two of five for Redis. On the Arm pair the free reading held on both brokers and the fitted shape failed on both (Figure S9). P1 and P9 need both readings. Load moved the fall toward shorter delivery times in every campaign, and P3a held in none.

TABLE S21

**ONE PREDICTION CONFIRMED ON EVERY CAMPAIGN, THREE ON ONE BROKER, AND THE REST NOT.** EVERY PRE-REGISTERED PREDICTION OF THE LAW CAMPAIGN, THE NUMBER IT WAS JUDGED ON (ALL RUNS; A SLOPE, RATIO OR MOVE WITH ITS 95% INTERVAL, FROM RESAMPLING WHOLE ROUNDS), AND ITS VERDICT, BROKER BY BROKER WHERE THE TWO DIFFER. A PREDICTION JUDGED BROKER BY BROKER IS CONFIRMED ONLY WHERE IT HOLDS ON BOTH. THE LAST COLUMN IS THE ONE VERDICT THAT MOVES WHEN THE FLAGGED, RECORDED OR PAUSED RUNS ARE LEFT OUT.

|       | Block, pair              | The prediction                                                                                                                               | On all runs                                                                                                                       | Verdict                 | Moves             |
|-------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------|
| P1    | L1, first x86            | the fall's halfway point inside $[s, s + h]$ at all but one of the slices a pair reaches, at least four of them; its slope on $s$ in 0.8–1.2 | slope, free and fitted: Kafka 0.758 (0.596–1.023) and 0.230; Redis 1.007 (0.844–1.169) and 0.662                                  | not confirmed           | –                 |
| P9    | L4, Arm                  | on the Arm pair, inside at every slice it reaches, at least two; the same slope                                                              | Kafka 0.830 (0.627–1.172) and 0.285; Redis 1.153 (0.753–1.325) and 0.798                                                          | not confirmed           | –                 |
| P2    | L2                       | the fall 2.5 to 5.5 times wider at a 4 ms tick than at 1 ms                                                                                  | ratio, free and fitted: Kafka 1.784 and 1.818; Redis 1.064 and 1.126 (S6.5)                                                       | not confirmed           | –                 |
| P2b   | L2                       | 6 to 14 times wider at a 10 ms tick                                                                                                          | Kafka 4.786 and 5.358; Redis 3.045 and 3.339 (S6.5)                                                                               | not confirmed           | –                 |
| P2c   | L2                       | the width on the tick a line through zero, slope 0.8–1.2                                                                                     | fitted slope: Kafka 0.968 (0.271–1.125); Redis 0.874 (0.807–1.034)                                                                | confirmed on Redis only | –                 |
| P2d   | L2                       | the fitted start within 0.25 ms of the slice at every tick                                                                                   | within it at one of twelve slice and tick pairs                                                                                   | not confirmed           | –                 |
| P3a   | L3, five Kafka campaigns | the halfway point moves less than 0.25 ms from 75% to 88% load                                                                               | free move: –0.009, –0.188, –0.086, –0.011 and –0.309 ms                                                                           | not confirmed on any    | first x86, 19 Sep |
| P3b   | L3, five Kafka campaigns | load raises the plateau                                                                                                                      | rise, free: 0.048 (0.044–0.052), 0.047, 0.048, 0.041 and 0.050                                                                    | confirmed on all five   | –                 |
| P4    | L7, first x86            | real-time priority cuts the plateau at least five times wherever it is at least 2%                                                           | Kafka 0.0405 to zero; Redis's plateau 0.0069 (S6.5)                                                                               | confirmed on Kafka only | –                 |
| P6    | L6                       | a recording of every thread's waits predicts the rate within 0.67–1.5                                                                        | median ratio 2.49 to 13.08 over four pair and broker parts; no setup inside                                                       | not confirmed           | –                 |
| P7    | L5, second x86           | the default slice follows the core count, and the fall follows it                                                                            | slices read back at 2, 4 and 8 CPUs as the rule gives; the fall on Kafka at 8 CPUs, Redis's fitted shape missing at all three     | confirmed on Kafka only | –                 |
| P8    | L8, first x86 and Arm    | Java has the same fall and priority effect, and Python's plateau is at most 1.5 times Java's                                                 | Python over Java: 3.251 (2.795–3.907), and 4.253 (4.013–4.560) on Arm                                                             | not confirmed           | –                 |
| L9-1  | L9, three pairs          | at least 90% of negative readings mostly spent waiting for a CPU                                                                             | 79.4 to 92.0%                                                                                                                     | not confirmed           | –                 |
| L9-2  | L9                       | the time-weighted waits predict the rate within 0.67–1.5                                                                                     | median ratio 0.104 to 0.165                                                                                                       | not confirmed           | –                 |
| L9-2b | L9                       | the share of acknowledgments whose own wait outlasted $D - g$ predicts it                                                                    | inside the rule on four of six parts; median ratio 1.11 to 1.30                                                                   | not confirmed           | –                 |
| M0    | M0, first x86            | four named causes of the delivery time's slope under bursts                                                                                  | slope on the added delay, traced half: Kafka 0.995; Redis 1.181 and 1.307; every cause dropped for Redis, two not tested on Kafka | –                       | –                 |

#### S6.4. Aberrant runs are marked, not removed, and one verdict moves without them

The plan keeps every run, and re-runs none whose conditions held. A program reads a mark for each aberration from the records, and an aberrant run carries its mark wherever it is drawn. Five marks keep a run out of every judge. They cover void runs, set aside before anything read them; the 18 runs of six false starts, each replaced at once by a campaign of the same design; and runs repeated, stopped by the lag brake, or ended before the integrity rule read them. Four marks leave a run counted: flagged, paused, far from its setup's other repeats, and traced by L9's or M0's recording. The brake would have stopped the 14 flagged runs, taken while it only recorded. In the 222 paused runs a message arrived more than 150 ms late, when the receiving side's disk stalled, for a cause not identified, or Kafka's broker could not reach its own controller. No run had a message arrive before it was sent. Without the flagged runs, or without all 1,110 taken while the brake only recorded, no verdict moves. Without the paused runs every verdict stands but P3a on the first x86 pair, whose fitted interval sits on the edge of its  $\pm 0.25$  ms bar: just outside it with them, just inside without them. The verdict on all the runs is the one reported. The postmortem, S36, counts every mark pair by pair, says why each was given, and draws, in four figures, all 4,823 runs of the law's blocks that measured a delivery time and a rate, of their 4,838.

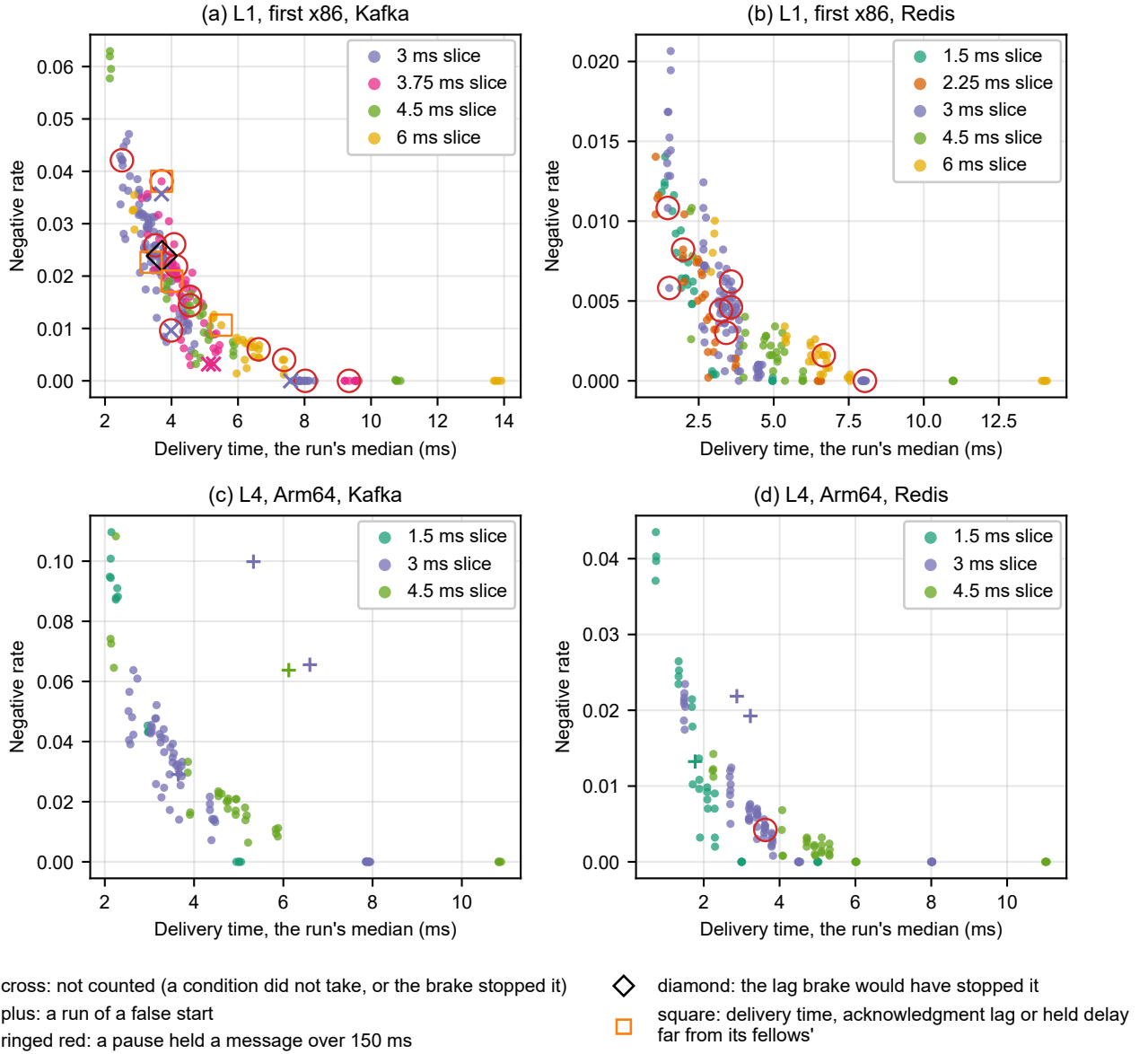


Fig. S9. **Slice runs fall from the shortest delivery time on, where the registered law put a plateau.** Every run of L1 (first x86 pair) and L4 (Arm pair, Arm64 in the panels) at its median delivery time and its negative-span rate, colored by the slice it ran at. Crosses: runs the integrity rule did not count. Plusses: runs of L4's two false starts. Rings, diamonds and squares: counted runs with a warning (S6.4). That the rate holds no plateau was read after the verdicts (S6.5).

### S6.5. Read after the verdicts: the wait is the rest of a slice, and two judges departed from the plan

Everything here is post hoc: it was read after the verdicts, and none of it changes one. The rate holds no plateau. At half the slice it is 1.45 to 3.94 times its value at 0.9 of the slice, in each of the ten slice and broker combinations whose runs reach half the slice. L9's recordings, at a 3 ms slice and a 1 ms tick, time the wait of the thread that timestamps each acknowledgment, for 1,516,342 acknowledgments in 304 runs. Of those whose thread waited more than 1 ms, 24.3 to 35.8% were still waiting at 3 ms and 3.3 to 6.8% past 4 ms, over twelve pair, broker and load parts. A whole-slice wait would keep all of them to 3 ms. A wait for the rest of the slice keeps 20% to 3 ms and none past 4 ms. The thread waits out what is left of the running task's slice,  $R$ , uniform on  $[0, s]$ , and then up to a tick,  $U$ , uniform on  $[0, h]$ , so that for  $h \leq s$

$$G(x) = \Pr[R + U > x] = \begin{cases} 1 - x^2/(2sh), & 0 \leq x \leq h, \\ 1 - (x - h/2)/s, & h \leq x \leq s, \\ (s + h - x)^2/(2sh), & s \leq x \leq s + h, \\ 0, & x \geq s + h. \end{cases} \quad (S5)$$

A thread preempted for a task starting its slice waits out all of it, but a thread woken while another task holds the processor waits out what is left. The acknowledgment's reply wakes the thread that timestamps it, so its wait is the second kind. A rate that falls from the start pulls a fitted plateau's fall below the slice and moves it less than the slice moves. That is the split between the free and fitted readings of P1 and P9 (postmortem, S16.10). The recording raised the rate it recorded, 1.36 to 1.49 times for Kafka and 3.69 to 4.33 times for Redis, so these waits describe the machines with it on.

In two places a frozen program departed from the plan, and in both the frozen answer is the one reported. The programs judging P2, P2b and P2c group L2's runs by tick alone, so each width they read spans the falls of both its slices, 1.5 and 3 ms. Fed Equation S4, the P2 program returns 1.98 free and 1.57 fitted for Redis, below its band, while one slice at a time the same formula gives 4.0 to 5.2 for P2 and 9.8 to 12.0 for P2b, inside both bands: those three verdicts carry no evidence about the law either way. P4's program words Redis "not confirmed" although its plateau, 0.69%, lies below the 2% at which P4 makes a claim, which the plan's rule reports as out of reach. What explains M0's slope on Redis was also found afterwards (postmortem, S37).

### S7. THE BROKER COMPARISON, GATED

Once the runs the sign check condemns are removed, Kafka and Redis are equivalent on the end-to-end latency, and a consumer setting that a co-located testbed hides decides their order.

The comparison is stated first on the end-to-end latency  $E$  of Equation 2, because  $E$  is a causal chain and the transport proxy  $S$  is not (Section II-A). Over the runs the sign check leaves standing,  $E$  is equivalent between the two brokers against a 40 ms margin, one broadcast frame at 25 frames per second, fixed three days before the campaign ran. The Hodges–Lehmann shifts are 0.682–0.859 ms, Kafka slower, and the highest upper bound of any 90% interval of the three estimators is 0.906 ms.  $E$  runs to  $t_{\text{out}}$ , so it also contains both clients' payload parse, which  $S$  and  $D$  carry for Kafka alone (Section VI). On  $S$  the two sit within a millisecond: a Hodges–Lehmann shift at one feed of 0.408 ms (90% bootstrap interval 0.389–0.419), with two one-sided tests (TOST) rejecting both nulls against a 1 ms margin at  $p < 0.001$ . The shift holds between 0.408–0.417 ms across three concurrency levels, and Welch's test, a percentile bootstrap and the Hodges–Lehmann estimator agree on equivalence at every level.

The two brokers are still distinguishable. The numbers on  $S$  come from a powered campaign that replays at a rate derived from the plan and verified against wall time, with 15 replicates at each of the three concurrency levels and a median of 125 events per run. It timestamps in microseconds on one clock, so Failure 2 cannot touch its numbers; Failure 1 can, and the sign check removed the runs it condemned (S2.2). On  $S$ , Kafka sits near 0.54 ms and Redis near 0.11 ms, and the 0.41 ms shift, Kafka slower, sits far from zero at every level, in the direction the gray-literature comparisons report. A second powered campaign, run ten hours later with the same protocol, reproduces the shift and fails the criterion registered for it, that its shift fall inside the first campaign's 90% interval at every level. Over the runs the check keeps it gives 0.381, 0.414 and 0.416 ms; the first, from the two Redis runs kept at one feed, falls 0.008 ms below the interval. We saw only after the result that the criterion was a poor test: a second estimate carries its own sampling error, so even identical shifts would satisfy it at all three levels only about 38% of the time. Restoring the runs the sign check condemned moves the shift by at most 0.016 ms (S2.2).

Part of even this shift is our harness. Timestamping Kafka's acknowledgment on the sending thread, as Redis's already is, with one request outstanding under matched load, shrinks the gap by 0.071 ms (95% bootstrap over runs: 0.039–0.113), and by 0.039 ms (95% bootstrap over runs: 0.008–0.059) in a replication at 30 runs per cell. What remains is a broker difference of 0.34–0.37 ms. The bias never turns a value negative, so the sign check, which rejects 40 of the 160 runs behind these numbers, cannot see it: sizing a bias of a few hundredths of a millisecond took a controlled experiment and a replication.

A consumption pattern reverses the order, and injected delay exposes it (Table S22). With one-way delay injected identically at both brokers, Redis's median end-to-end latency reaches 4,651 ms at 5 ms of delay and 103,143 ms at 50 ms, against Kafka's 12.4 and 336.6 ms. Redis's transport proxy grows about 930 to 2,050 times the delay applied, while Kafka's tracks it almost additively. The cause is the consumption pattern, not the broker. Our Redis consumer acknowledges each entry and waits for the reply, the pattern the tutorials teach, so a read of two hundred entries is followed by two hundred sequential round trips. Each trip is free at 0.22 ms and hopeless at 20 ms, where the service rate falls below the arrival rate and backlog sets the delay. Kafka's consumer serves from a prefetched buffer and commits on a timer, so it pays the delay once per batch.

TABLE S22

**PER-MESSAGE ACKNOWLEDGMENT TURNS INJECTED DELAY INTO SECONDS OF BACKLOG FOR REDIS ALONE. MEDIAN END-TO-END LATENCY  $E$  AND TRANSPORT PROXY  $S$  UNDER ONE-WAY DELAY INJECTED IDENTICALLY AT BOTH BROKERS, FIVE FEEDS PER CONFIGURATION. THE ZERO-DELAY ROW IS REJECTED IN BOTH CONFIGURATIONS, SO THERE IS NO USABLE BASELINE.**

| Delay | Kafka $E$ | Kafka $S$ | Redis $E$  | Redis $S$  |
|-------|-----------|-----------|------------|------------|
| 0 ms  | rejected  |           |            |            |
| 5 ms  | 12.4 ms   | 5.6 ms    | 4,651 ms   | 4,645 ms   |
| 20 ms | 77.8 ms   | 20.8 ms   | 31,401 ms  | 31,268 ms  |
| 50 ms | 336.6 ms  | 44.9 ms   | 103,143 ms | 102,460 ms |

TABLE S23

EVERY CELL OF OUR FIRST RESULT SET FAILS THE SIGN CHECK. THE TRANSPORT PROXY  $S$  (MEDIAN, MS) BY CONCURRENCY  $N$ , WORKSTATION TESTBED,  $10\times$  REPLAY, WITH THE HOLM-CORRECTED  $p$  OF EACH COMPARISON.

| $N$ | Kafka | Redis | Difference | $p_{\text{Holm}}$     |
|-----|-------|-------|------------|-----------------------|
| 1   | 0.999 | 1.006 | 0.007      | $1.0 \times 10^{-4}$  |
| 5   | 1.001 | 1.197 | 0.196      | $6.5 \times 10^{-6}$  |
| 10  | 1.002 | 1.346 | 0.344      | $9.0 \times 10^{-11}$ |

The falsifier was recorded before the run. If the acknowledgment pattern is the cause, batching the acknowledgments must remove most of the effect without touching the broker, the network or the injected delay. At one feed and 20 ms of delay, batching them two hundred at a time cut the median end-to-end latency from 4,138 ms to 103 ms, a factor of 40, replicated four times. Read-loop instrumentation shows each read returning every waiting entry, a median of 106, in about 32 ms, so the constraint sits in the acknowledgment path. At five feeds batching left the backlog in place, 92.4 s batched against 87.6 s unbatched, and that null is unexplained (postmortem, S19).

A co-located testbed cannot see any of this. The round trip the two patterns multiply is a few hundred microseconds there, so paying it once per event or once per batch differs by less than the noise. The setting that decides the comparison under any realistic separation is therefore invisible where most comparisons are run. That is the shape of the paper’s two failures: a configuration decides the result, and nothing in the output records it.

## S8. RESULTS WE WITHDREW

The sign check condemned our own first result, which had passed every conventional check we knew. A second early result, a twentyfold end-to-end gap between the brokers, rested on Kafka runs that all passed the sign check and still measured only the harness starting up. Table S25 lists every other result we withdrew or withheld.

### S8.1. The first result set passed every conventional check

An invalid corpus can look decisive. On the workstation testbed we replayed real matches at ten times real speed over one, five and ten feeds, with 15–30 runs per cell. The transport proxy separated the brokers cleanly: every comparison was significant after correction, and at  $N \geq 5$  the run distributions did not overlap (Table S23). The sign check of Section II-D rejects 862 of the testbed’s 1,382 runs, 109 of the 126 behind the table among them, so none of the table’s 6 conditions is usable; 8 of the testbed’s 76 conditions survive. The table reports  $S$ , which our first harness timed from the acknowledgment, and  $S$  is negative whenever the acknowledgment lag exceeds the delivery time (Equation 3).

The corpus was not naive. Five earlier corrections had shaped it, each found by investigating a result that looked wrong and each reversing the apparent winner:

- 1) A process-relative clock. Its per-process origin carried each run’s consumer start-up offset into every measurement.
- 2) A replay speed of  $120\times$ , at which the producer could not keep pace.
- 3) An asymmetric load generator. The Kafka producer waited a broker round trip per event, and pipelining it cut median end-to-end latency at one feed from 1,644 ms to 16 ms.
- 4) A concurrency axis that was covertly a throughput axis.
- 5) A heavy tail in every mean. The end-to-end 95th percentile of Kafka sat at 99–105 ms at every concurrency, and decomposition placed it in send lag.

A fairness audit found payloads and serialization costs comparable. A sixth problem was an option the orchestration script accepted and never passed on, which turned the first run of the acknowledgment experiment into a null. Since then every intervention carries a manipulation check that aborts the campaign if the treatment is not accepted.

A corpus that had survived six corrections, a fairness audit, and a full battery of rank tests, effect sizes and multiplicity correction had none of its conditions usable, and nothing in its output said so.

### S8.2. The twentyfold end-to-end gap was send lag at the kick-off

A corpus can pass the sign check and still measure its own start-up. On the cloud testbed the historical comparison E1 put the brokers’ median end-to-end latencies near 105 and 5 ms. Of the 105.5 ms for Kafka, 102.9 ms was send lag, while the transport proxy was 0.84 and 0.80 ms for the two brokers. Our first analysis reported the gap as a finding, attributed it to the client’s sender thread waking from idle, and built a practitioner recommendation on it. E1’s Kafka runs all passed the sign check, and its four cells’ medians agreed within half a millisecond, 105.3 to 105.7 ms.

That send lag is paid once per run. Replaying one match at a verified real-time rate under three observation windows, all else fixed, we counted the events that woke more than 50 ms late (Table S24). As the events emitted grow from 57 to 507, that count stays at four and exactly one send blocks, so the share paying the cost falls from 7.0% to 0.8%, as  $1/\text{events}$ : the

TABLE S24

**COUNT OF LATE EVENTS DOES NOT GROW WITH THE WINDOW. SAME MATCH, DRIVER AND BROKER, ONE FEED, TRUE REAL-TIME REPLAY; ONLY THE OBSERVATION WINDOW CHANGES. BOTH PRODUCERS ARE LOOP-TRACED. COUNTS ARE MEDIAN OVER THREE REPLICATES FROM THE PER-EVENT TRACE; THE SEND-LAG PERCENTILES COME FROM THE PER-RUN SUMMARY.**

|       | Window<br>(s) | Events  |         | Send lag (ms) |       | Events > 50<br>ms late | Blocking<br>sends |
|-------|---------------|---------|---------|---------------|-------|------------------------|-------------------|
|       |               | emitted | matched | median        | max   |                        |                   |
| Kafka | 60            | 57      | 57      | 1.56          | 103.4 | 4                      | 1                 |
|       | 180           | 148     | 148     | 1.61          | 103.5 | 4                      | 1                 |
|       | 600           | 507     | 465     | 1.58          | 103.5 | 4                      | 1                 |
| Redis | 60            | 57      | 57      | 1.50          | 17.4  | 0                      | 0                 |
|       | 180           | 148     | 148     | 1.56          | 20.4  | 0                      | 0                 |
|       | 600           | 507     | 465     | 1.53          | 24.8  | 0                      | 0                 |

dilution a once-per-run cost predicts. A per-event constant would have grown the count with the window. The median send lag of Kafka stays at 1.6 ms throughout, and its maximum at 103.5 ms.

The per-event trace shows why. The first send blocks for 102.6 ms, and the single-threaded replay loop wakes each event due meanwhile about 103 ms late. A Kafka producer fetches metadata before its first send to a topic it has not yet cached [S26]. The shape matches, although we did not isolate the call. Every plan opens with two lineup records, two half-start markers and the kick-off pass, all at second zero, so in the plan behind the sweep the first five events share one timestamp. The first pays the block and the other four wait it out. A football feed therefore delivers its densest burst when the producer is coldest. The Redis producer has no such prologue, because XADD to a new stream creates it in the same round trip.

E1 matched a median of seven events per run. A median of seven values at 102.93 ms needs at least four of them that late, and the trace shows exactly four per run, so the matched sample was almost entirely prologue. The argument holds at every replay rate E1 could have used (postmortem, S3). The twentyfold figure is withdrawn, and the broker comparison now rests on powered campaigns that match a median of 125 events per run (Section VI). A percentile over single-digit samples is a statement about the harness.

## S9. RELATED WORK IN FULL

Clock resolution and scheduling delay are each well understood, and rules of the form we propose already exist in neighboring fields. What the literature lacks is their consequence for the sub-millisecond comparisons that broker benchmarks now publish, whose tools discard the evidence that would show it.

### S9.1. Benchmark variability is structural, and here it has a mechanism

Run-to-run instability in benchmarks is structural, and the literature that shows it supports our reading. Mytkowicz et al. showed that factors as innocuous as link order and environment-variable size flip published conclusions [S27], and Georges et al. that JVM results without confidence intervals routinely mislead [S28]. Barrett et al. found that virtual machines frequently never reach the steady state their benchmarks assume, so the distribution being sampled is itself non-stationary [S29]. That is the closest published analogue of the mobility we see between identical runs. Maricq et al. found hundreds of trials necessary for stable comparisons [S30]. Against this stands only common practice: the surveyed norm of a handful of repetitions presumes a stable distribution [S31], [S32]. What we add is the mechanism. Where the send interval is commensurate with the timestamp resolution, the instability is not variance around a value. It is a two-point support whose weights move between runs, so no replicate count converges (Section IV-C).

Tail latency under load is established, and what is new is the measurement it corrupts. Li et al. decompose the tail latency of three Linux servers and find measured tails substantially worse than a queueing model alone predicts [S33]. The excess comes from background processes, request re-ordering, interrupt routing, power management and NUMA effects. We add that the stall distribution propagates into the measurement of an unrelated quantity, and that load moves its tail far faster than its core. From idle to the knee of the load sweep the negative-span rate grows  $61\times$  and the jitter's width  $5\times$  (S3.4). A benchmark can therefore sit at a utilization where its measurements look tight, because the jitter is narrow, while the tail that corrupts them is already growing.

### S9.2. The resolution literature knew the problem, and the deletion is new

The problem behind Failure 2 is older than every tool Section V reads. Danzig and Melvin built microsecond timers for workstations in 1990 because the available clock was too coarse for the interval being timed [S34]. The arithmetic is older still. Hewlett-Packard's Application Note 162-1 carries the retention identity, the reduced fraction the grid turns on, the ceiling on resolution gain, the symptom and the cure [S6]. The one thing it does not contain is the failure this paper reports, because

TABLE S25

EVERY OTHER RESULT WE WITHDREW OR WITHHELD, AND WHAT DEFEATED IT. ONE ROW PER RESULT, WITH THE SECTIONS OF THE POSTMORTEM THAT TELL IT IN FULL.

| What we claimed                                                                                                                                   | Why it fell                                                                                                                                                                                                                                                                                      | Postmortem   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| The benchmark's discards were causality violations.                                                                                               | Separated by sign, the Kafka-driver corpus's 10,913,263 discards hold not one negative. They are resolution failures.                                                                                                                                                                            | S1.3         |
| A candidate mechanism for the Redis-driver negatives: skew between two views of a virtualized clock.                                              | The driver takes the publish time from the entry identifier the Redis server assigns. Its span is a difference between two millisecond-floored clocks on two hosts, for which $-1000\ \mu\text{s}$ is the only negative value expressible.                                                       | S1.3         |
| Failure 1 was a causality violation.                                                                                                              | The acknowledgment and the receipt are two branches of one cause. Recounted, $S$ is negative on 62,264 of 738,730 events and the send-referenced span on 0.                                                                                                                                      | S1.4         |
| An M/G/1 waiting time, then our own registered bracket, described how the rate grows with utilization.                                            | Over $\rho = 0.881\text{--}0.990$ M/G/1 fitted worse than the mean ( $R^2 = -0.05$ ), and the bracket predicted $2.45\text{--}3.07\times$ growth against $1.44\times$ measured. Real-time priority showed that $\rho$ is not the variable.                                                       | S1.5, S5.1   |
| A traced tail index confirmed the payload sweep's exponent.                                                                                       | Two proper estimators on the same histogram disagree sixfold (0.21 and 1.19). The agreement was a coincidence of window and estimator.                                                                                                                                                           | S1.6, S8     |
| Six predictions registered before the comprehensive campaign, and a jitter-kernel conjecture after it.                                            | One prediction was confirmed, three falsifiers fired, one became unevaluable and one split. An alternation test killed the conjecture.                                                                                                                                                           | S1.7, S1.15  |
| The achieved send rate, for part of the campaign.                                                                                                 | It was copied from the configuration flag, which for a subset of runs disagreed with the rate against elapsed wall time. The affected conditions were re-derived from per-event timestamps.                                                                                                      | S1.8         |
| A 2026 comparison gave Redis a p99 of 0.8 ms against Kafka's 12.5 ms.                                                                             | Those figures are not on the page cited. They belong to another comparison that a search returned beside it; the postmortem cites both.                                                                                                                                                          | S1.9         |
| The benchmark's cross-host exposure, measured by running it.                                                                                      | Its distributed mode failed on every attempt, across two releases. The exposure rests on its source and our own two-host harness.                                                                                                                                                                | S1.10, S22.1 |
| The injected-delay sweep (H1): a rank correlation across 0–50 ms of delay at the broker supported the rate's fall as the delivery time lengthens. | In the re-run at one feed the delay was common-mode and canceled in $S$ : over 0–20 ms end-to-end latency tracked it, 3.72 to 23.61 ms, while $S$ stayed flat.                                                                                                                                   | S3.1         |
| A recorded replay rate for every run we report.                                                                                                   | Plans carry a built-in $120\times$ compression, so the speed-up flag is not the rate, and for the earliest corpora, E1 among them, no surviving artifact records the achieved one. For E1 the start-up cost reads the rate back as true real time, and the original script, found later, agrees. | S4           |
| A co-location campaign, meant to shorten the delivery time.                                                                                       | Moving the broker onto the driver lengthened $S$ at idle, 0.512 to 0.573 ms, and moved it $1.03\times$ under load. Nothing was manipulated, so nothing was tested.                                                                                                                               | S4.1         |
| The two brokers were equal on the transport proxy, from E1.                                                                                       | E1 matched seven events per run. The powered runs' first seven events reproduce its near-equality; over all events Redis is 0.41 ms faster.                                                                                                                                                      | S2, S5       |
| Configurations whose send interval over the timestamp resolution has an even denominator ( $q = 2, 4$ ) could not test the grid.                  | An on-grid configuration is predicted flat, so they are evidence for the phase account. The exclusion also flipped under a rate-local estimate of the continuous value.                                                                                                                          | S6           |
| At $q = 5$ , retention near 40% (registered).                                                                                                     | The median was 51.04%. The replicates split about 40% and 60%, the two branches the phase account requires.                                                                                                                                                                                      | S6           |

its counter keeps both readings and uses the retained fraction as its estimator. Deleting the population that carries the fractional part turns a solved metrology problem back into an open one.

The mathematics is a century old. Weyl's equidistribution theorem [S35] governs the incommensurate regime of the send interval over the timestamp resolution in the long run, and the three-distance theorem [S36] the phases a finite run of sends visits. Schuchman's condition [S37], generalized by Wannamaker et al. [S38], underlies the dithering that Table IV recommends. The timing literature treats a coarse timestamp as staleness, not deletion.

Coordinated omission is the closest neighbor, and its mirror image [S39]. There the worst samples are never taken, so percentiles are optimistic by orders of magnitude. Under Failure 2 the samples are taken and the latency is computed, and a downstream filter rejects it. One loses the slow tail, the other the fast bulk, and a correction for one does nothing about the other. The benchmark we instrument records into an HdrHistogram, which ships coordinated-omission machinery and none for this failure [S40]. Its sound rejection of negative values is what motivated the positivity filter, so a protection against one measurement error created the conditions for another.

What we claim as new is narrow. Quantization is elementary, and Cloudprofiler exists because milliseconds are known to be too coarse [S41]. The new element is the interaction of a quantized timestamp, a producer with few send phases and a positivity filter. It yields a retained fraction that is bimodal, irreproducible between identical runs and invisible in the reported summary, with its damage set by the lowest-terms denominator of the send interval over the timestamp resolution.

TABLE S26  
**RULES OF THE RIGHT FORM ALREADY EXIST, IN NEIGHBORING FIELDS.** EACH ADDRESSES THE TIMESTAMP RESOLUTION OR THE SAMPLES A MEASUREMENT CANNOT USE; NONE IS APPLIED WHERE A BROKER BENCHMARK REPORTS A LATENCY.

| Source                                     | What it requires or does                                                                                                                                       | How far it reaches                                                                                     |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| CoreMark-PRO run rules [S42]               | Each workload runs for at least 1000 times the minimum timer resolution.                                                                                       | A floor relative to the resolution. Plain CoreMark sets an absolute 10 s floor instead.                |
| SPEC power methodology, clause 8.2.2 [S43] | Should report the samples that may return “Unknown” and those exceeding the maximum allowed uncertainty.                                                       | Counts unusable readings; a “should”, in a methodology document.                                       |
| OWAMP [S44]                                | Records a packet timestamped in the future unaltered, within its timeout window.                                                                               | Keeps an impossible value as data.                                                                     |
| ETSI TS 102 250-4 [S45, Clause 5.1]        | Requires removed measurement cycles to be reported.                                                                                                            | Counts what was removed.                                                                               |
| RFC 7679 [S46, Sec. 3.5]                   | Treats negative singletons as usable distribution data.                                                                                                        | Keeps the sign.                                                                                        |
| TimeWeaver [S47]                           | Keeps measurements with missing timestamps or negative latencies, in lower accuracy tiers. States that the prior method rejected about 3,631 of 8,804 clients. | Retention with a quality label, and a published rejection rate. Network measurement, counting clients. |

### S9.3. Rules of the right form exist, in other fields

Standardized benchmarking already contains rules of the form we propose. CoreMark-PRO requires each workload to run for at least 1000 times the minimum timer resolution [S42]. That is a floor on the measured interval relative to the timestamp resolution, the ratio Equation 4 turns on. Network measurement goes further and keeps the impossible value as data (Table S26).

The flagship standardized benchmark for message-oriented middleware has none of this. SPECjms2007 records delivery time in a histogram of 100 ms default granularity [S48], [S49]. Its run rules, design document and user’s guide contain no resolution clause, no minimum measurable interval and no counter of discarded latency samples. Their only timing-quality requirement, clause 3.2, bounds inter-node clock agreement at 100 ms, which is a skew limit. The guide does discard throughput samples, by configured constants, but it never counts the latency samples its own resolution makes unusable.

The nearest prior sighting of a negative span inside a benchmark was blamed on a clock. Klenik and Pataricza report negative derived durations in a Hyperledger Fabric benchmark, call them a serious violation of event causality, and trace them to a drifting peer clock [S50]. The durations oscillate smoothly around zero, a drift signature, which puts them in the channel of our Redis-driver  $-1000\ \mu\text{s}$  negatives and outside Failure 1. Their check is post-hoc data validation, with no per-run gate and no campaign rate.

### S9.4. The run queue inverts spans that a backlog cannot

A span can invert with no backlog at all, and that is where our reading parts from the concurrent report. Sharma et al. count negative timing spans in distributed inference systems and propose a `causality_health` check [S51]. Injecting skew, they observe no violations up to 3 ms and clear ones by 5 ms, and read that as a skew threshold. We do not dispute their measurements, only one sentence of the reading: “queueing alone cannot produce negative timing spans”.

Under the queue they model, a backlog at a service stage, the sentence is true, and Section III refutes that account of our own corpus as well. As written it is false, because two queues are in play. A message waits in a service backlog; the thread that reads the clock waits in the kernel’s run queue for a processor. The second wait delays a timestamp whether or not any backlog exists, and our rates are highest at high utilization with no backlog at all.

If a backlog cannot invert a span, removing the backlog leaves skew as the only explanation, and a millisecond threshold on skew becomes a sufficient check. On a loaded machine it is not. On a single host, where one clock leaves no inter-host offset at all, we see a negative-span rate of 30.5% at 88% utilization, and a scheduling parameter moves it fiftyfold without touching the clock (Table II). Our cross-host clocks cannot settle the question either way. Between our hosts `chrony` reports an offset of  $67\ \mu\text{s}$ , far inside the 3 ms at which Sharma et al. saw no violations, but it bounds its own error at 4–7 ms per host. We therefore claim no cross-host bound (Section VIII), and the single-host rate needs none.

The skew reading is older than this dispute. The benchmark’s maintainer gave it in 2021, attributing publish latency above end-to-end latency to inter-node skew on a report from a single node [S52]. It does not consider whether the thread that took the timestamp was running.

Their own formulation is nearer ours than the dichotomy allows. They note that the minimum separation between the two events “exhibits variability due to scheduling jitter”, and that violations grow likelier as the skew exceeds its quantiles. That is the condition of Equation 3 with skew written where we write the stall. A threshold in milliseconds of skew cannot transfer to a machine where the stall is the term that grows.

### S9.5. Production tracing reached the conclusion without the mechanism

Production tracing concluded, from the symptom alone, that rewriting an inverted span’s timestamps does harm. Distributed tracing systems ship clock-skew adjusters that “modif[y] start time and log timestamps for spans that appear to be ‘off’ with

respect to the parent span” [S53]. An issue titled “Clock Skew Adjuster considered harmful” called the result “wrong data and really hard to diagnose” [S54], and Jaeger has defaulted the adjustment off since 2020. A July 2026 bug against the OpenTelemetry eBPF instrumentation reports child spans starting before their parents on same-node resources, from a reporter who had checked for clock skew and found none [S55]. Its deltas of seconds to minutes are orders of magnitude beyond any run-queue stall, so their cause is not Failure 1. The report shows the absence of a second hypothesis: a practitioner who has ruled out the clock, on one node, has nowhere else to look. If the negative span comes from a late timestamp, an adjuster moves evidence instead of correcting an error, by an amount Section III measures.

### S9.6. A runnable thread does not always run

Work-conservation violations leave runnable threads waiting while cores sit idle. Lozi et al. document Linux scheduler bugs that do so for seconds, invisible to conventional testing because throughput stays healthy [S56]. Those are CFS-era defects. Our kernel’s scheduler is EEVDF, and we make no claim that they are present.

Queueing gives the leading-order account of waiting, and it is right about what matters here: utilization alone does not fix a waiting time, because pooling and geometry matter at fixed  $\rho$ . The geometry contrast of Section III-B therefore refutes a single-parameter law and leaves multi-server queueing, which is geometry-dependent at fixed  $\rho$ , standing. Chandrasekar and Kramberger apply such a single-parameter M/G/1 framing to benchmark bias [S57]. We adopted it too, and on the load axis it fitted worse than the mean (Table S25). On a second axis they reach half of our mechanism, concurrently and independently: their client’s own interpreter lock “artificially inflates” the latencies it reports, charging the harness’s scheduling to the server under test. That is the inflation half. What a positive number cannot show remains ours: the negative span, and the sign channel that shows the acknowledgment was timestamped late.

Li et al. show real-time priority collapsing a tail that renicing leaves alone, on the system under test [S33]. We apply the same lever to the harness (Section III-B). An agent acting on a lagged observation is a problem of partial observability, studied in this journal for load balancing under delayed feedback [S58]. There the lag is a known property of the system. Here it sits inside the benchmark, unknown to those reading its output.

## REFERENCES

- [S1] StatsBomb, “Statsbomb open data,” *GitHub Repository*, 2023. [Online]. Available: <https://github.com/statsbomb/open-data>
- [S2] M. Mohammad, “Analysis of design patterns and benchmark practices in Apache Kafka event-streaming systems,” *arXiv preprint arXiv:2512.16146*, 2025.
- [S3] Linux Kernel Documentation, “KVM-specific MSRs: MSR\_KVM\_SYSTEM\_TIME\_NEW,” <https://docs.kernel.org/virt/kvm/x86/msr.html>, 2026, accessed: Aug. 21, 2026.
- [S4] Z. Zhou, “sched: Reduce the default slice to avoid tasks getting an extra tick,” Linux kernel mailing list, commit 2ae891b82695, 2025.
- [S5] S. Bradner, “Benchmarking terminology for network interconnection devices,” Internet Engineering Task Force, RFC 1242, 1991.
- [S6] Hewlett-Packard, “Time interval averaging,” Hewlett-Packard Company, Application Note 162-1, 1970, part no. 02-5952-0766.
- [S7] J. H. Gliever and D. C. Chu, “Measurement averaging counting apparatus employing a randomly phase modulated time base to improve counting resolution,” U.S. Patent 3,938,042, Feb., 1976, assigned to the Hewlett-Packard Company. Accessed: Sep. 11, 2026.
- [S8] IEEE, “IEEE standard for terminology and test methods for analog-to-digital converters,” IEEE Std 1241-2010, 2011.
- [S9] Joint Committee for Guides in Metrology, “Evaluation of measurement data — guide to the expression of uncertainty in measurement (JCGM 100:2008),” [https://www.bipm.org/documents/20126/2071204/JCGM\\_100\\_2008\\_E.pdf](https://www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf), 2008, accessed: Aug. 20, 2026.
- [S10] J. W. Pratt, “Remarks on zeros and ties in the Wilcoxon signed rank procedures,” *Journal of the American Statistical Association*, vol. 54, no. 287, pp. 655–667, 1959.
- [S11] Apache Software Foundation, “Apache Kafka tools — EndToEndLatency.java and ProducerPerformance.java,” Source repository, 2026, <https://github.com/apache/kafka>. Accessed: Aug. 31, 2026.
- [S12] Apache Pulsar contributors, “PerformanceConsumer.java,” Source repository, 2026, <https://github.com/apache/pulsar>. Accessed: Aug. 21, 2026.
- [S13] Synadia, “NATS cli — cli/latency\_command.go,” Source repository, 2026, <https://github.com/nats-io/natscli>. Accessed: Aug. 31, 2026.
- [S14] OpenMessaging Project, Linux Foundation, “The OpenMessaging benchmark framework,” 2018, available at [openmessaging.cloud/docs/benchmarks/](https://openmessaging.cloud/docs/benchmarks/). Accessed: Aug. 19, 2026.
- [S15] Broadcom, “RabbitMQ PerfTest — Consumer.java,” Source repository, 2026, <https://github.com/rabbitmq/rabbitmq-perf-test>. Accessed: Aug. 31, 2026.
- [S16] IOP Systems, “Rezolus: systems performance telemetry — scheduler\_discarded\_samples,” Source repository, 2026, <https://github.com/iopsystems/rezolus>. Accessed: Aug. 20, 2026.
- [S17] A. D. Brunelle, “blktrace/btt — btt/inlines.h,” Source repository, 2026, accessed: Aug. 20, 2026.
- [S18] EMQ Technologies, “emqtt-bench: MQTT benchmark tool — src/emqtt\_bench.erl,” Source repository, 2026, <https://github.com/emqx/emqtt-bench>. Accessed: Aug. 21, 2026.
- [S19] J. Axboe, “fio: Flexible I/O tester — gettime.c,” Source repository, 2026, <https://github.com/axboe/fio>. Accessed: Aug. 20, 2026.
- [S20] G. Tene, “wrk2 — src/wrk.c,” Source repository, 2026, <https://github.com/giltene/wrk2>. Accessed: Aug. 21, 2026.
- [S21] Apache Kafka contributors, “KAFKA-19888: Clamp negative values in coordinator histograms,” Issue tracker and pull request #20986, 2025, <https://issues.apache.org/jira/browse/KAFKA-19888>. Accessed: Aug. 21, 2026.
- [S22] —, “Kafka Streams StreamTask.java: end-to-end latency recording (KIP-613),” Source repository, 2026, <https://github.com/apache/kafka/blob/trunk/streams/src/main/java/org/apache/kafka/streams/processor/internals/StreamTask.java>. Accessed: Sep. 13, 2026.
- [S23] S. Guo, “Avoid adding negative metric values (pull request #56),” OpenMessaging Benchmark, commit ebb5d6b8, Mar. 2018, <https://github.com/openmessaging/benchmark/pull/56>. Accessed: Aug. 18, 2026.
- [S24] Confluent, “librdkafka introduction — delivery reports,” <https://github.com/confluentinc/librdkafka>, 2026, accessed: Aug. 20, 2026.
- [S25] J. Axboe, “fio HOWTO — interpreting the output,” <https://fio.readthedocs.io/>, 2026, accessed: Aug. 20, 2026.
- [S26] Apache Kafka, “KIP-526: Reduce producer metadata lookups for large number of topics,” Apache Software Foundation, Kafka Improvement Proposal, 2020.

- [S27] T. Mytkowicz, A. Diwan, M. Hauswirth, and P. F. Sweeney, “Producing wrong data without doing anything obviously wrong!” in *Proc. 14th Int. Conf. Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. ACM, 2009, pp. 265–276.
- [S28] A. Georges, D. Buytaert, and L. Eeckhout, “Statistically rigorous Java performance evaluation,” in *Proc. 22nd ACM SIGPLAN Conf. Object-Oriented Programming, Systems, Languages and Applications (OOPSLA)*. ACM, 2007, pp. 57–76.
- [S29] E. Barrett, C. F. Bolz-Tereick, R. Killick, S. Mount, and L. Tratt, “Virtual machine warmup blows hot and cold,” *Proceedings of the ACM on Programming Languages*, vol. 1, no. OOPSLA, pp. 52:1–52:27, 2017.
- [S30] A. Maricq, D. Duplyakin, I. Jimenez, C. Maltzahn, R. Stutsman, and R. Ricci, “Taming performance variability,” in *Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. USENIX Association, 2018, pp. 409–425.
- [S31] E. van der Kouwe, G. Heiser, D. Andriesse, H. Bos, and C. Giuffrida, “SoK: Benchmarking flaws in systems security,” in *Proc. 4th IEEE EuroS&P*. IEEE, 2019, pp. 310–325.
- [S32] T. Hoefler and R. Belli, “Scientific benchmarking of parallel computing systems: Twelve ways to tell the masses when reporting performance results,” in *Proc. Int. Conf. High Performance Computing, Networking, Storage and Analysis (SC)*. ACM, 2015, pp. 73:1–73:12.
- [S33] J. Li, N. K. Sharma, D. R. K. Ports, and S. D. Gribble, “Tales of the tail: Hardware, OS, and application-level sources of tail latency,” in *Proc. ACM SoCC*. ACM, 2014.
- [S34] P. B. Danzig and S. Melvin, “High resolution timing with low resolution clocks and a microsecond resolution timer for sun workstations,” *ACM SIGOPS Operating Systems Review*, vol. 24, no. 1, pp. 23–26, Jan. 1990.
- [S35] H. Weyl, “Über die gleichverteilung von zahlen mod. eins,” *Mathematische Annalen*, vol. 77, pp. 313–352, 1916.
- [S36] V. T. Sós, “On the distribution mod 1 of the sequence  $n\alpha$ ,” *Annales Universitatis Scientiarum Budapestinensis, Sectio Mathematica*, vol. 1, pp. 127–134, 1958.
- [S37] L. Schuchman, “Dither signals and their effect on quantization noise,” *IEEE Transactions on Communication Technology*, vol. 12, no. 4, pp. 162–165, 1964.
- [S38] R. A. Wannamaker, S. P. Lipshitz, J. Vanderkooy, and J. N. Wright, “A theory of nonsubtractive dither,” *IEEE Transactions on Signal Processing*, vol. 48, no. 2, pp. 499–516, 2000.
- [S39] G. Tene, “How NOT to measure latency,” 2015, talk, Strange Loop / QCon.
- [S40] —, “HdrHistogram: A high dynamic range histogram,” 2015, available at [hdrhistogram.org](https://hdrhistogram.org).
- [S41] S. Yang, J. Jeong, B. Scholz, and B. Burgstaller, “Cloudprofiler: TSC-based inter-node profiling and high-throughput data ingestion for cloud streaming workloads,” *arXiv preprint arXiv:2205.09325*, 2022.
- [S42] EEMBC, “CoreMark-PRO run rules,” <https://github.com/eembc/coremark-pro>, 2026, accessed: Aug. 20, 2026.
- [S43] Standard Performance Evaluation Corporation, “SPEC power and performance benchmark methodology, version 2.3,” [https://www.spec.org/power/docs/SPEC-Power\\_and\\_Performance\\_Methodology.pdf](https://www.spec.org/power/docs/SPEC-Power_and_Performance_Methodology.pdf), 2025, accessed: Aug. 20, 2026.
- [S44] S. Shalunov, B. Teitelbaum, A. Karp, J. W. Boote, and M. J. Zekauskas, “A one-way active measurement protocol (OWAMP),” Internet Engineering Task Force, RFC 4656, Sep. 2006.
- [S45] ETSI, “Speech and multimedia transmission quality (STQ); QoS aspects for popular services in mobile networks; part 4: Requirements for quality of service measurement equipment,” European Telecommunications Standards Institute, Technical Specification TS 102 250-4 V2.3.1, Nov. 2019.
- [S46] G. Almes, S. Kalidindi, M. Zekauskas, and A. Morton, “A one-way delay metric for IPPM,” Internet Engineering Task Force, RFC 7679, Jan. 2016, obsoletes RFC 2679.
- [S47] R. Durairajan, S. K. Mani, P. Barford, R. Nowak, and J. Sommers, “TimeWeaver: Opportunistic one way delay measurement via NTP,” in *Proc. 30th International Teletraffic Congress (ITC 30)*, 2018, pp. 185–193.
- [S48] Standard Performance Evaluation Corporation, “SPECjms2007 run and reporting rules, version 1.0,” <https://www.spec.org/jms2007/docs/RunRules.html>, 2014, accessed: Aug. 20, 2026.
- [S49] K. Sachs, S. Kounev, J. Bacon, and A. P. Buchmann, “Performance evaluation of message-oriented middleware using the SPECjms2007 benchmark,” *Performance Evaluation*, vol. 66, no. 8, pp. 410–434, 2009.
- [S50] A. Klenik and A. Pataricza, “Adding semantics to measurements: Ontology-guided, systematic performance analysis,” *Acta Cybernetica*, 2022, arXiv:2112.11270.
- [S51] A. Sharma, D. Shah, D. Lariviere, and H. ElBakoury, “Time, causality, and observability failures in distributed AI inference systems,” *arXiv preprint arXiv:2604.21361*, Apr. 2026, Open Compute Project, Unified Intelligent Infrastructure workstream.
- [S52] OpenMessaging Benchmark contributors, “Can the average publish latency be larger than end-to-end latency? (issue #216),” <https://github.com/openmessaging/benchmark/issues/216>, 2021.
- [S53] Jaeger contributors, “model/adjuster/clockskew.go,” <https://github.com/jaegertracing/jaeger>, 2026, version 1.57.0. Accessed: Aug. 21, 2026.
- [S54] —, “Clock skew adjuster considered harmful (issue #1459),” Issue tracker, 2019, <https://github.com/jaegertracing/jaeger/issues/1459>. Accessed: Aug. 21, 2026.
- [S55] OpenTelemetry eBPF Instrumentation contributors, “Child span start time precedes parent span start time (issue #2568),” Issue tracker, 2026, <https://github.com/open-telemetry/opentelemetry-ebpf-instrumentation/issues/2568>. Accessed: Sep. 13, 2026.
- [S56] J.-P. Lozi, B. Lepers, J. Funston, F. Gaud, V. Quéma, and A. Fedorova, “The Linux scheduler: a decade of wasted cores,” in *Proc. 11th European Conf. Computer Systems (EuroSys)*. ACM, 2016, pp. 1–16.
- [S57] A. Chandrasekar and J. Kramberger, “Identifying and mitigating systemic measurement bias in production LLM inference benchmarks,” *arXiv preprint arXiv:2605.24217*, 2026.
- [S58] A. Tahir, B. Alt, A. Rizk, and H. Koepl, “Load balancing in compute clusters with delayed feedback,” *IEEE Transactions on Computers*, vol. 72, no. 6, pp. 1610–1622, 2023.